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Murdoch et al.

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(54) **SECURING AUTHENTICATION FLOWS
USING A DECENTRALIZED IDENTIFIER**

(58) **Field of Classification Search**

None

See application file for complete search history.

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This patent is subject to a terminal dis-
claimer.

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(57) **ABSTRACT**

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A digital wallet generates an identification value associated with a DID of a DID owner. The digital wallet generates a first request including the identification value for an authentication token from an identification provider. The first request is provided to the identification provider. The digital wallet receives, in response to the identification provider validating the first request, the authentication token that authenticates the digital wallet with a verifiable claim issuer including the identification value from the identification provider. The digital wallet generates a second request for one or more verifiable claims from the verifiable claim issuer. The second request includes the DID and authentication token including the identification value. In response to the verifiable claim issuer validating the authentication token and the identification value, one or more verifiable claims from the verifiable claim issuer are received by the digital wallet.

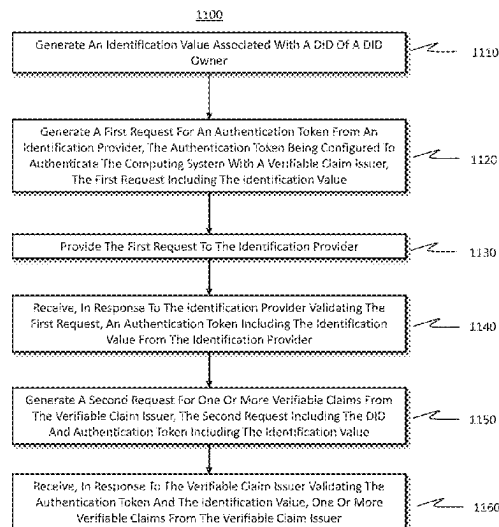
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Mar. 31, 2022, now Pat. No. 12,021,868.

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(Continued)

(52) **U.S. Cl.**
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(2013.01); **H04L 9/38** (2013.01); **H04L 9/50**
(2022.05)

20 Claims, 12 Drawing Sheets



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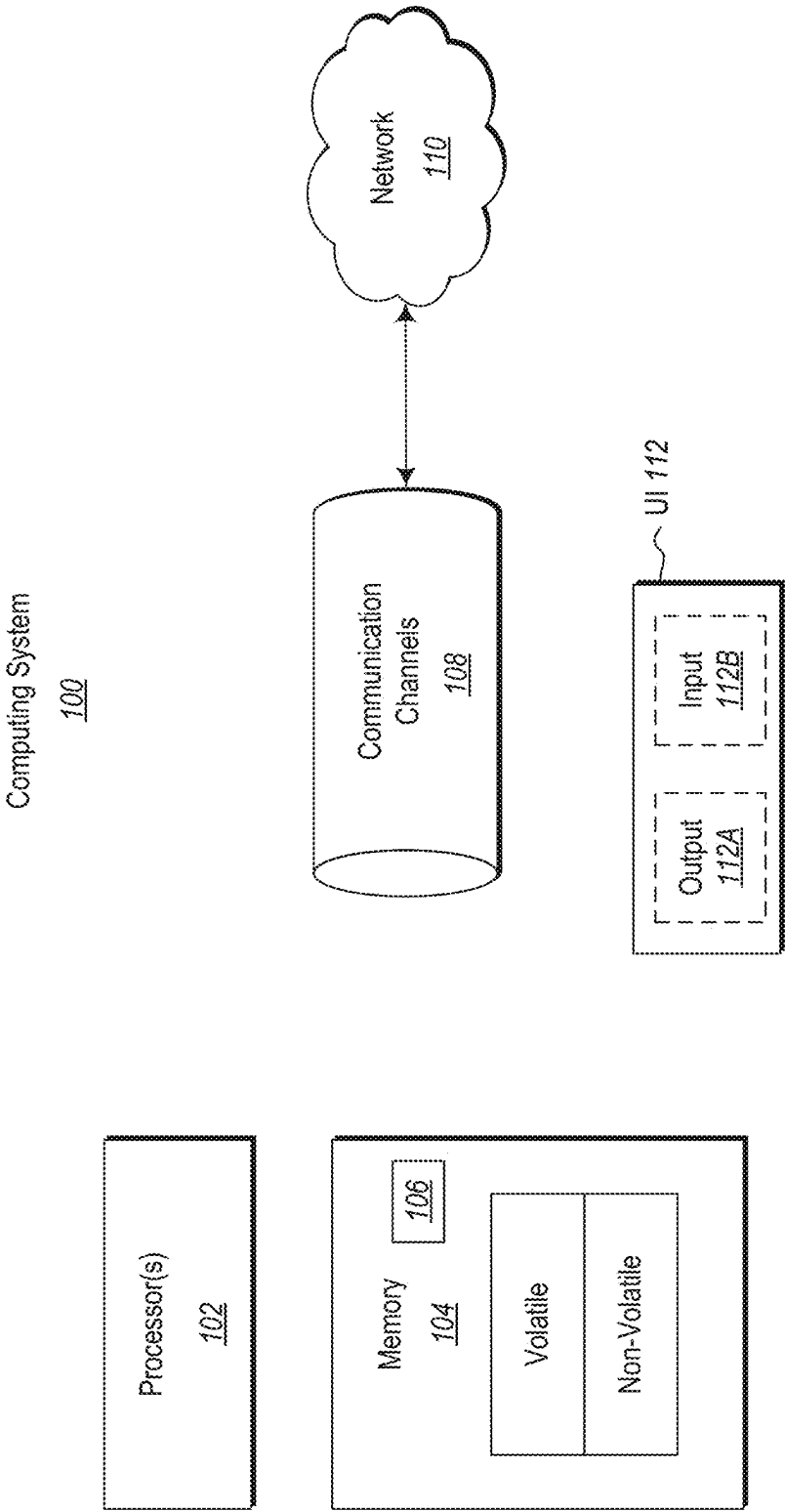


FIG. 1

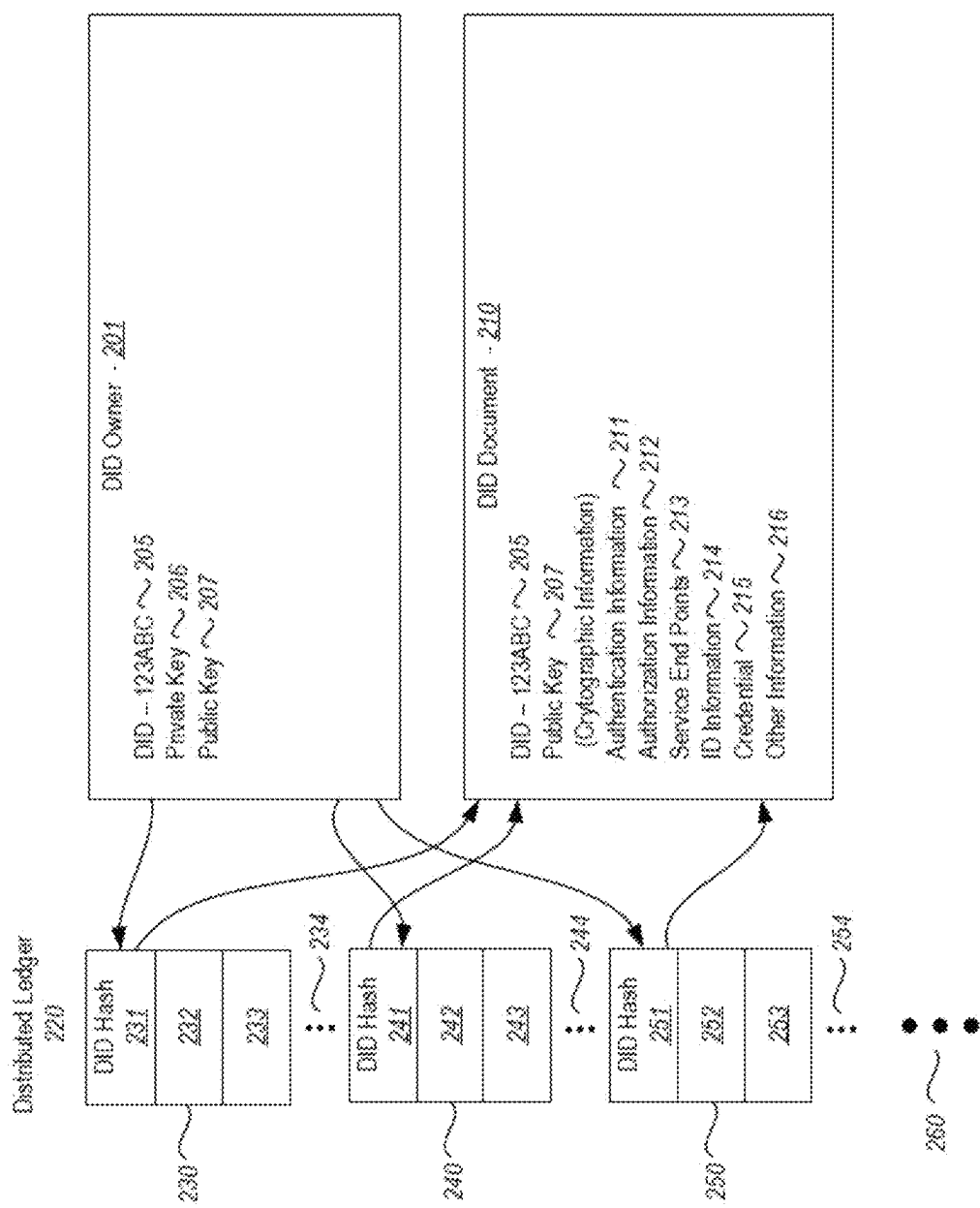


FIG. 2

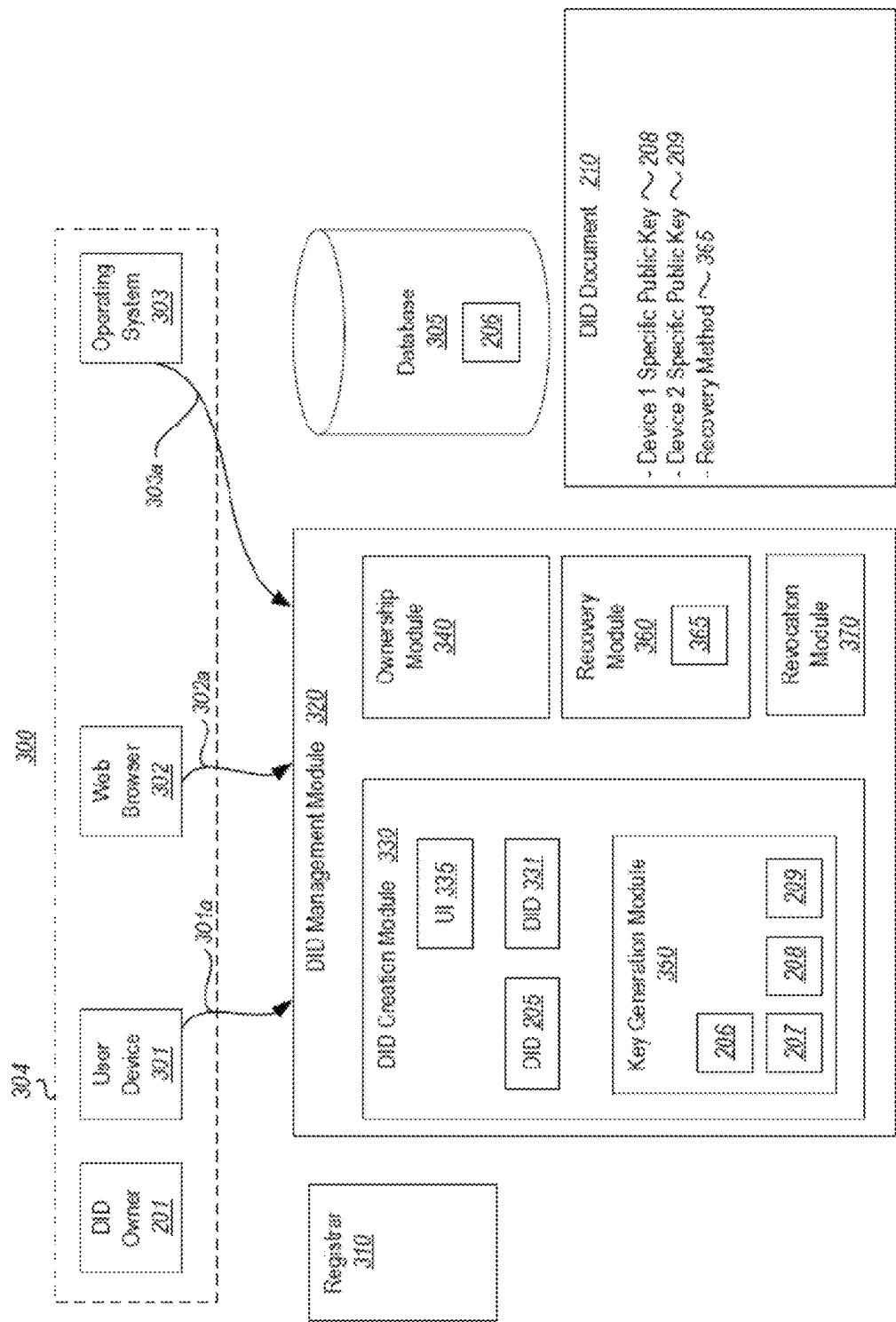


FIG. 3

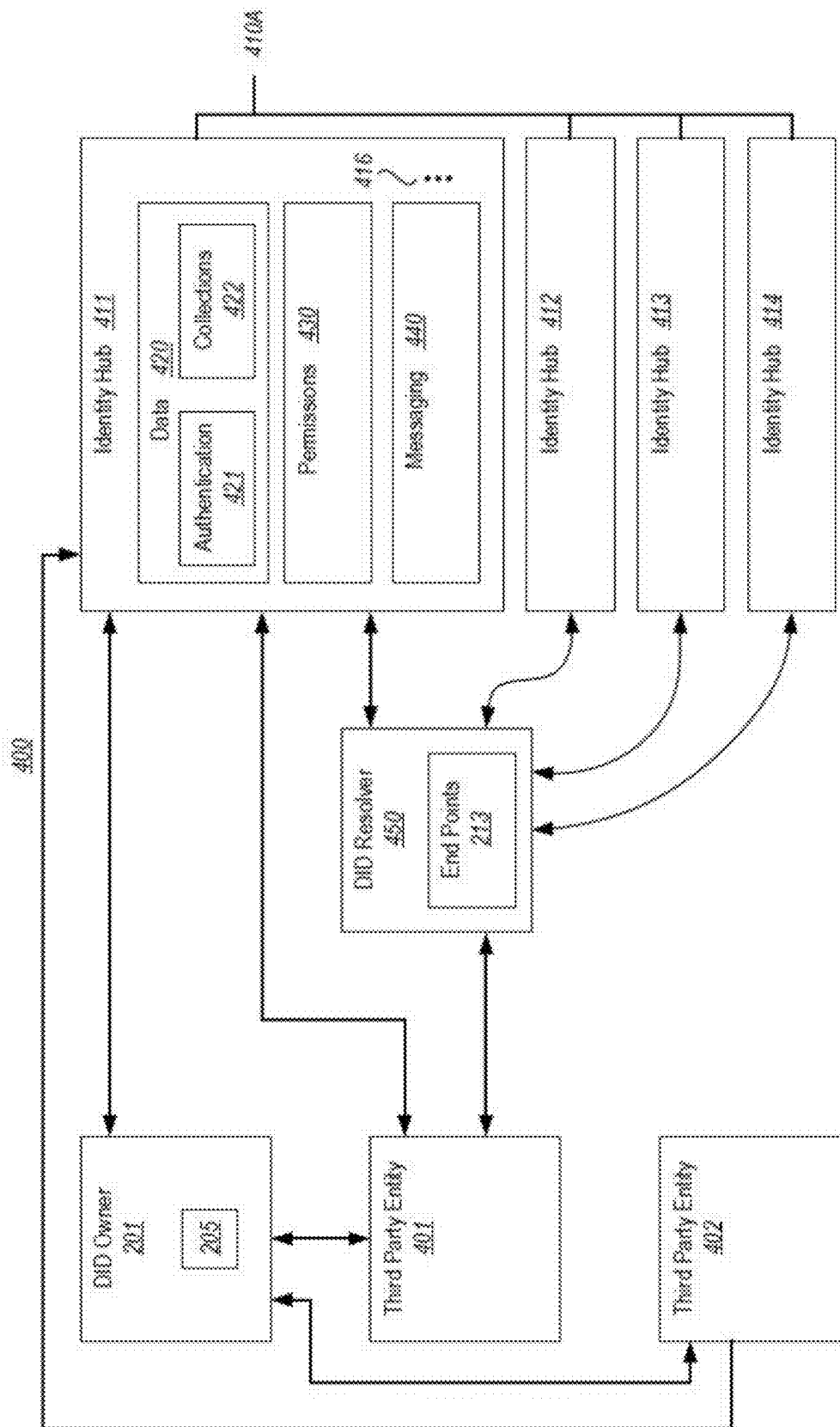


FIG. 4

500

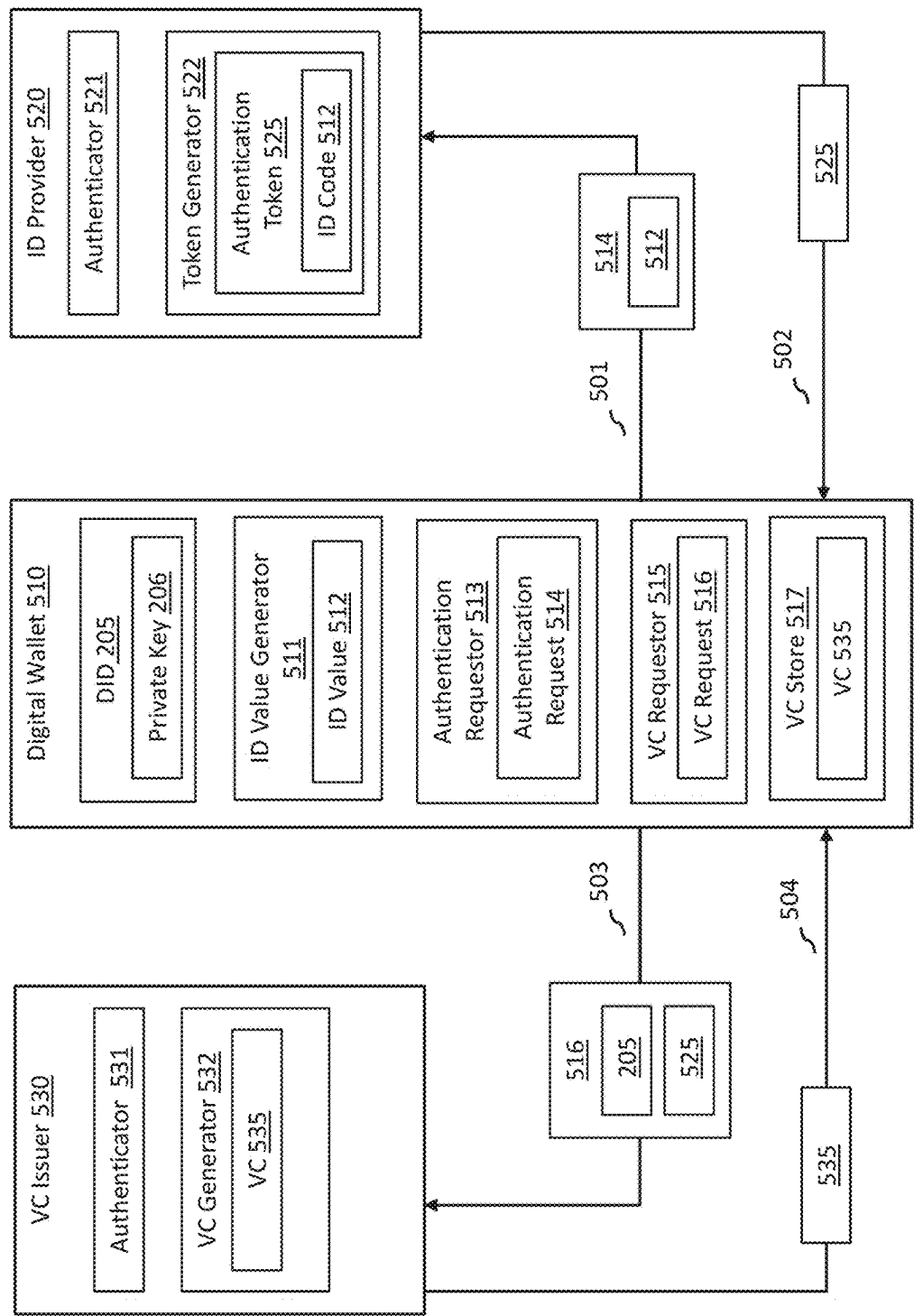


FIG. 5

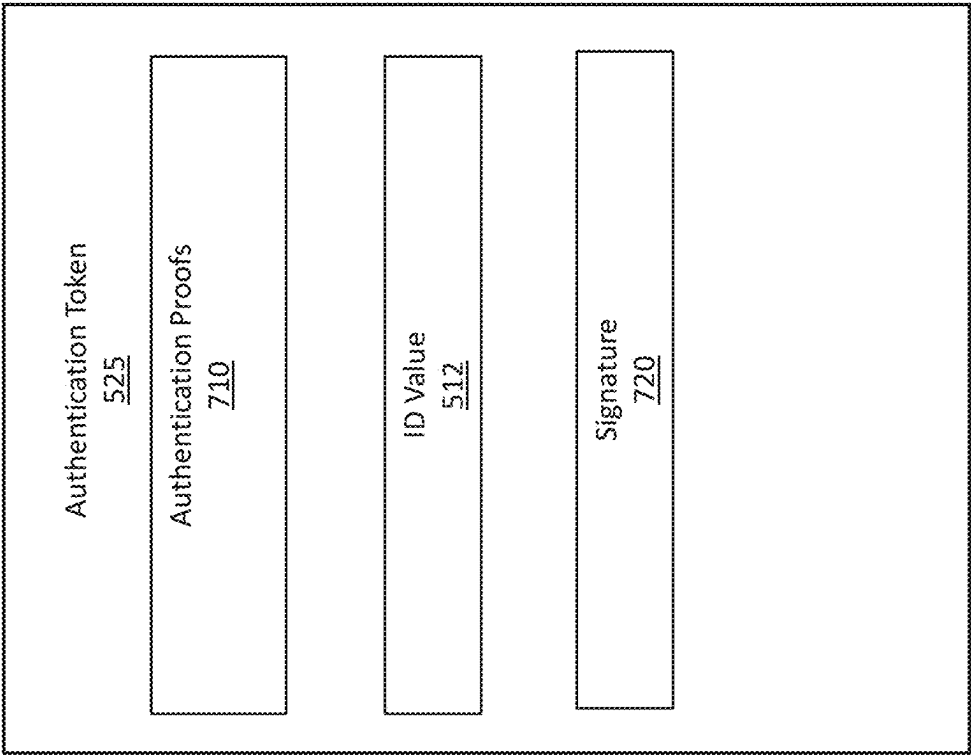


FIG. 7

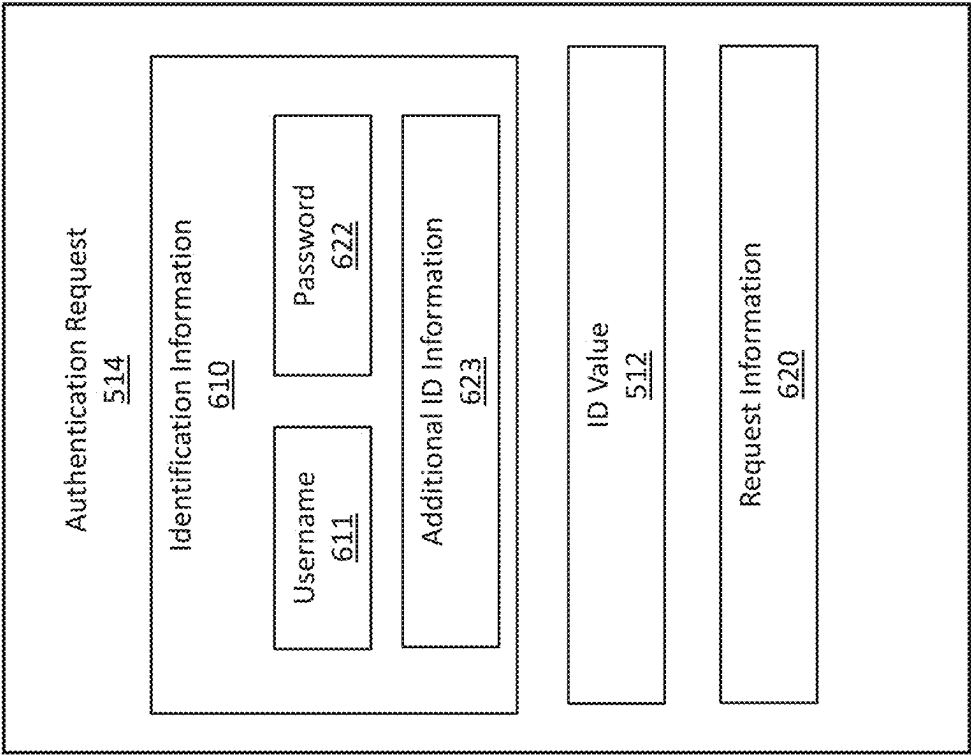


FIG. 6

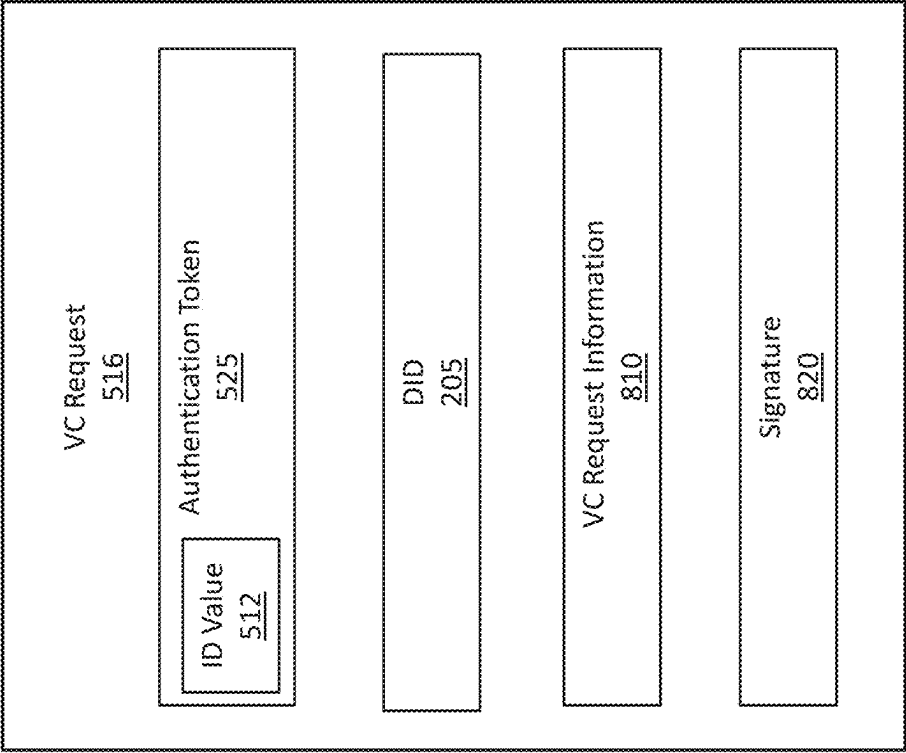


FIG. 8

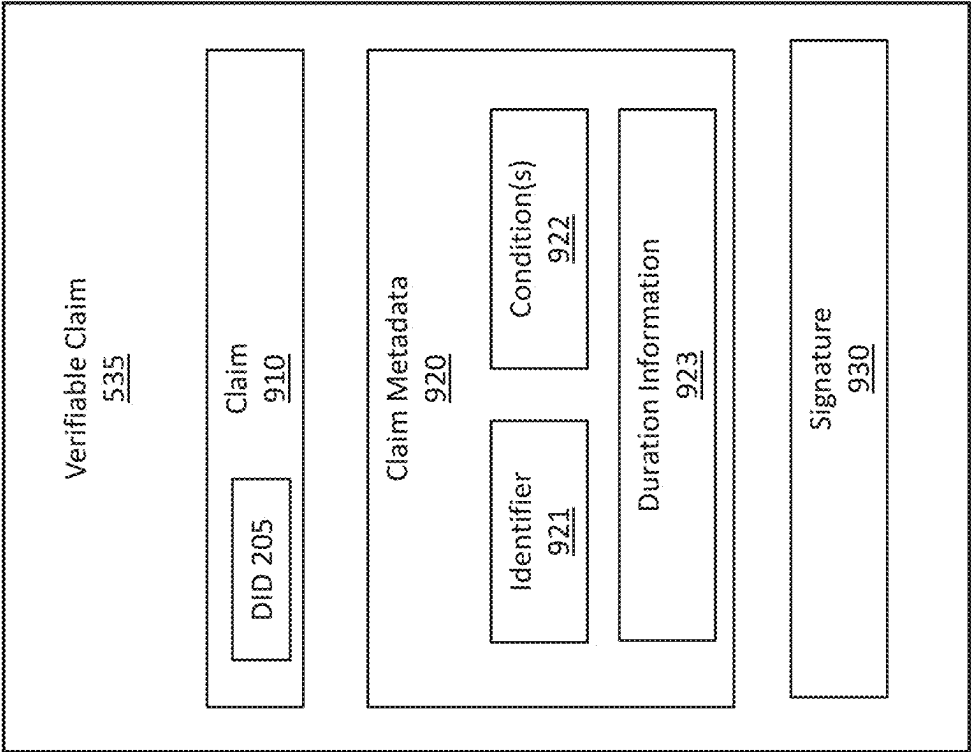


FIG. 9B

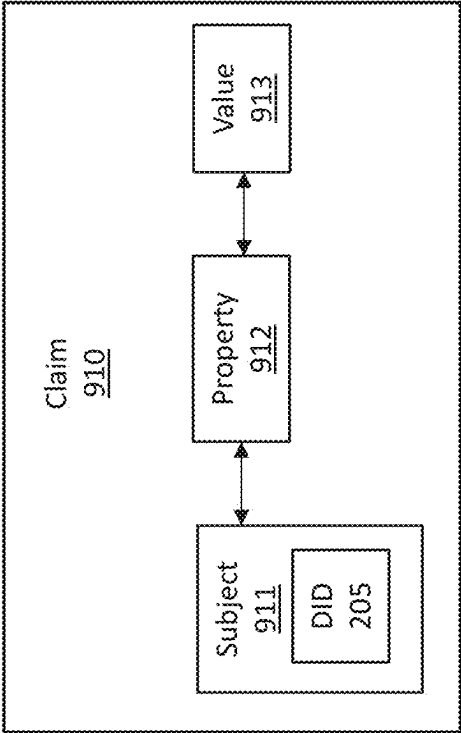


FIG. 9A

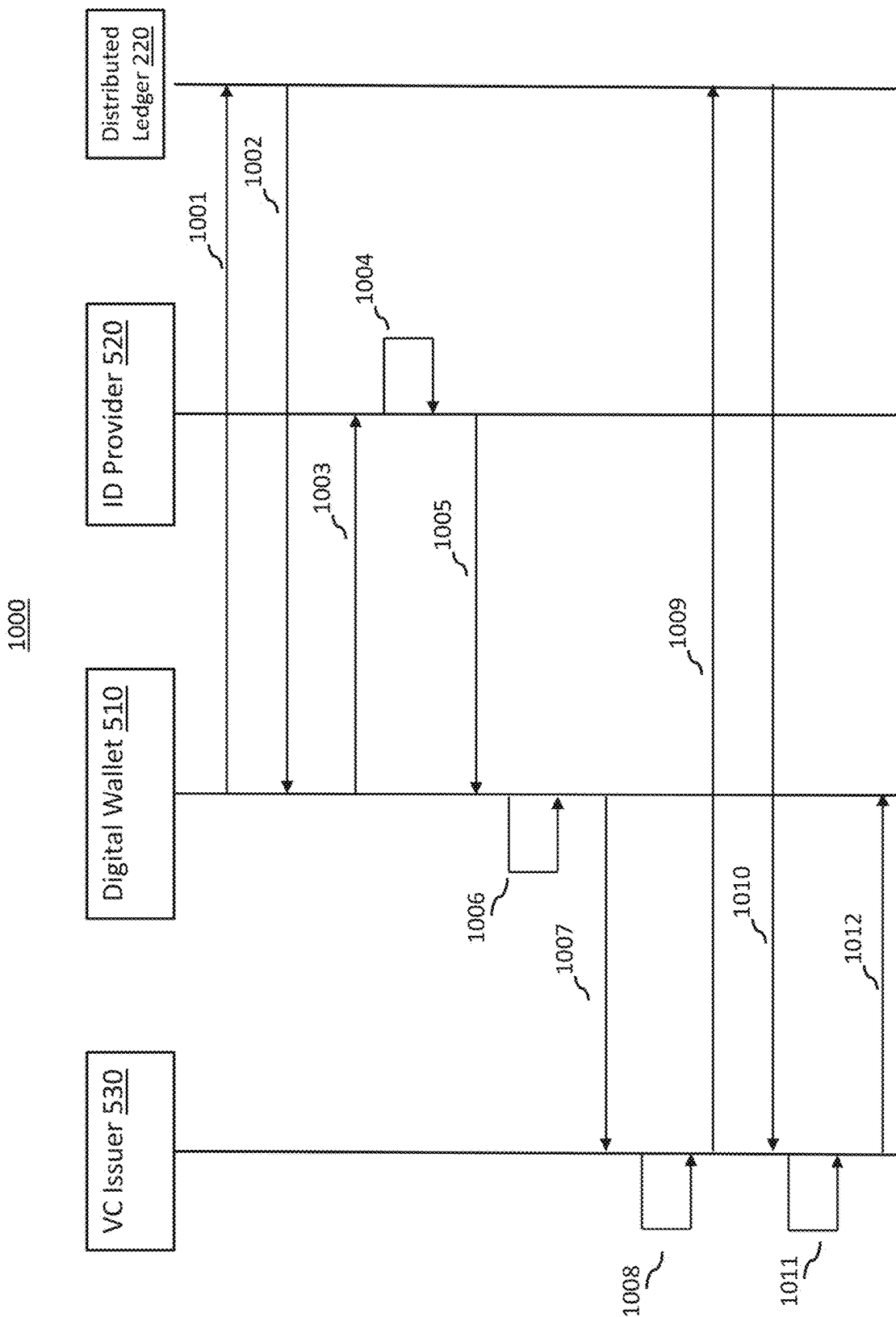
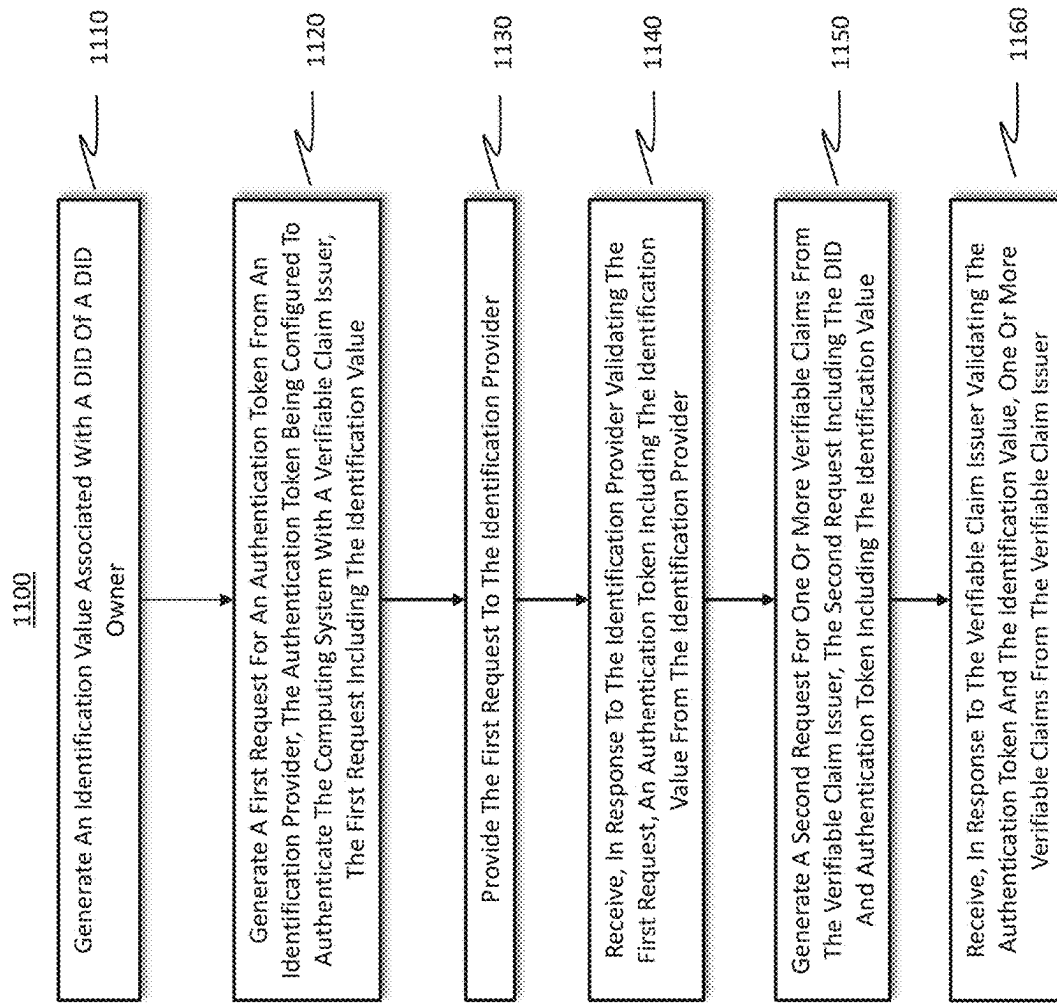


FIG. 10

**FIG. 11**

1200

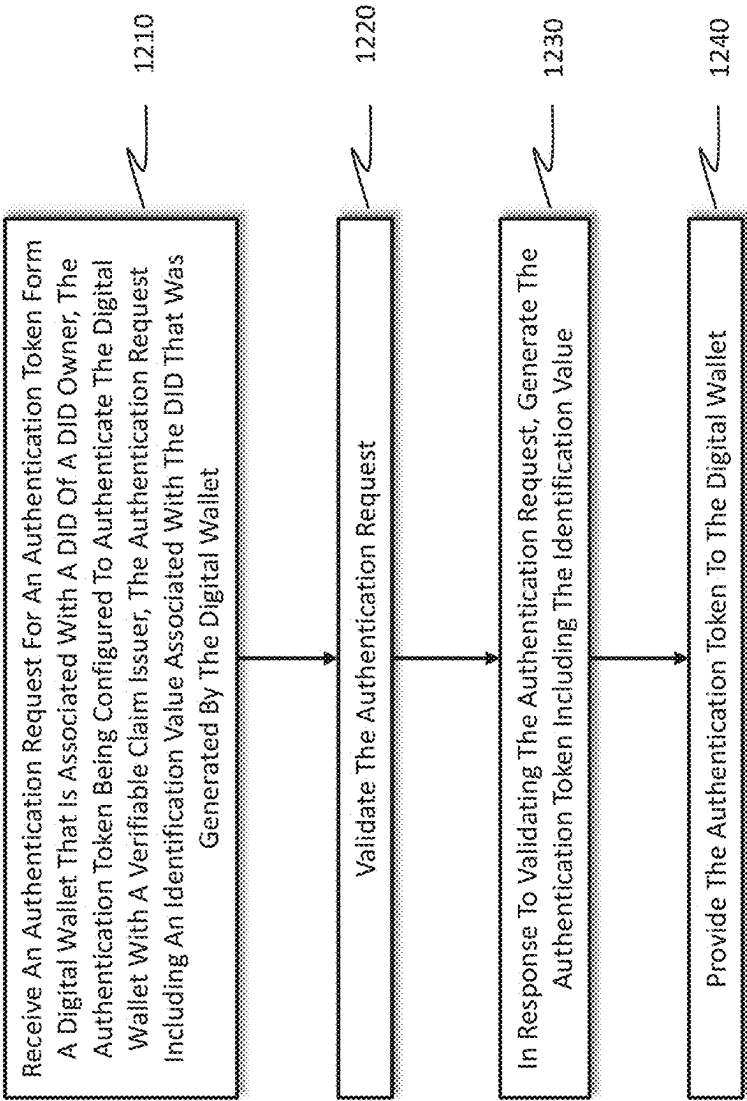
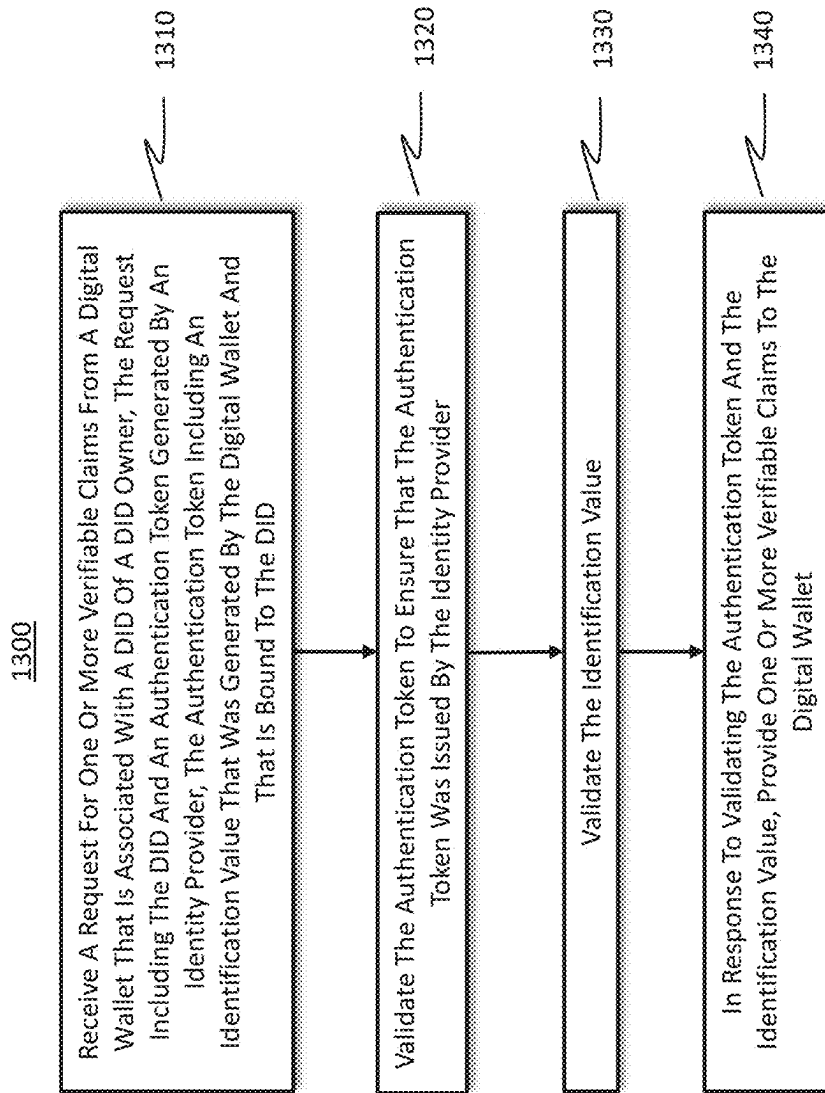


FIG. 12

**FIG. 13**

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SECURING AUTHENTICATION FLOWS USING A DECENTRALIZED IDENTIFIER

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 17/710,220 filed Mar. 31, 2022, the disclosures of which is incorporated herein by reference in its entirety.

BACKGROUND

A digital identity is a mechanism to keep track of an entity across different digital contexts. After an identity is determined, appropriate action can be taken related to the entity that has the identity. As an example, authorizations, privileges, customizations, and access can be provided to the entity. Thus, digital identities are an important mechanism to ensure that information is restricted to appropriate trust boundaries via appropriate containment of authorizations and privileges. Digital identities are also an important mechanism to ensure a positive and consistent user experience when accessing their data and customizations.

Most currently used documents or records that prove identity are issued by centralized organizations, such as governments, corporations, schools, employers, or other service centers or regulatory organizations. These organizations often maintain every member's identity in a centralized identity management system. A centralized identity management system is a centralized information system used for organizations to manage the issued identities, their authentication, authorization, roles, and privileges. Centralized identity management systems have been deemed as secure since they often use professionally maintained hardware and software. Typically, the identity issuing organization sets the terms and requirements for registering people with the organization. When a party needs to verify another party's identity, the verifying party often needs to go through the centralized identity management system to obtain information verifying and/or authenticating the other party's identity.

Decentralized Identifiers (DIDs) are a more recent type of identifier. Decentralized identifiers are independent of any centralized registry, identity provider, or certificate authority. Distributed ledger technology (such as blockchain) provides the opportunity for using fully decentralized identifiers. Distributed ledger technology uses distributed ledgers to record transactions between two or more parties in a verifiable way. Once a transaction is recorded, the data in the section of ledger cannot be altered retroactively without the alteration of all subsequent sections of ledger. This provides a fairly secure platform in which it is difficult or impossible to tamper with data recorded in the distributed ledger. Since a DID is generally not controlled by a centralized management system, but rather is owned by an owner of the DID, DIDs are sometimes referred to as identities without authority.

The subject matter claimed herein is not limited to embodiments that solve any disadvantages or that operate only in environments such as those described above. Rather, this background is only provided to illustrate one exemplary technology area where some embodiments describe herein may be practiced.

BRIEF SUMMARY

This Summary is provided to introduce a selection of concepts in a simplified form that are further described

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below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

5 Computing technologies provide for a data structure called a "verifiable claim or credential". In these technologies, a verifiable claim issuer makes one or more claims about a subject and generates the verifiable claim. The verifiable claim includes those claim(s) as well as proof instructions to prove that claim(s) have not been tampered with and were indeed issued by the verifiable claim issuer. The verifiable claim also often includes duration information metadata that defines a period of time that the verifiable claim is valid for use or that defines a specific number of times that the verifiable claim is authorized for use. In a decentralized environment, the verifiable claim also includes a Decentralized Identifier (DID) of the verifiable claim issuer. The verifiable then provides the verifiable claim to a claims holder (i.e., the subject of the claim made in the verifiable claim), for presentation to any relying party that relies upon the veracity of those claims.

As an example, the verifiable claim issuer might be a computing system associated with a government agency in charge of issuing driver licenses. The government agency computing system may generate a verifiable claim with claims about a citizen, such as the birthdate, residence address, weight, eye color, hair color, authorization to drive, restrictions on authorization to drive, and so forth. The government agency computing system issues the verifiable claim to the citizen. If the citizen is stopped by law enforcement, a computing system of the citizen may present the verifiable claim, whereby a computing system associated with law enforcement may use the proof instructions to verify that the claims were issued by the government agency and indeed have not been tampered with since issuance. In another example, a computing system of an organization that provides inoculations may issue claims to a parent of a child that assert that the child has received certain inoculations. The computing system of the parent may then present these inoculation claims to a school where the child is to attend. In the above examples, the relying party was the law enforcement agency and the school the child attends or more specifically, the computing system of the law enforcement agency and the school.

To obtain a verifiable claim from a verifiable claim issuer, the subject of a claim often makes a request to the verifiable claim issuer to obtain a verifiable claim that can later be presented to a relying party. For example, a citizen may request that the government agency provide him or her a verifiable claim related to the driver's license, or the parent may request the organization that provides the inoculations provide him or her with a verifiable claim related to their child's inoculation history. In response to the request, the verifiable claim issuer will provide the verifiable claim to the subject of the claim once any procedures related to the issuance of the verifiable claim have been satisfied.

In many instances, however, the verifiable claim issuer may not be sure that the entity that makes a request for a verifiable claim is actually the subject of the claim. For example, it is possible that a malicious party may try to obtain verifiable claims related to the subject of the claim so as to pass off the verifiable claims as his or her own verifiable claims. For instance, a parent who does not believe in inoculations may spoof a request to the organization that provides inoculations so as to pretend to be the claim subject. In this way, the parent who does not believe in the inoculations may receive a verifiable claim intended

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for the child of the claim subject and then pass this claim off for their own child. Thus, the school would believe the child of the malicious parent was inoculated when in fact he or she may not be.

The embodiments presented herein provide a novel solution to the above discussed problem. The embodiments presented herein utilize a DID of a DID owner who desires to request one or more verifiable claims from a verifiable claim issuer. A digital wallet owned by the DID owner generates an identification value that binds the identification value to the DID of the DID owner. The identification value may be a nonce value or randomized hash value that is signed using the private key, a hash of the DID, or even the DID itself.

The embodiments presented herein also utilize an identity provider. The identity provider is an entity that is configured to provide authentication and validation for the verifiable claim issuer of entities that wish to receive one or more verifiable claims from the verifiable claim issuer. Thus, the identity provider is a trusted source from the perspective of the verifiable claim issuer and the verifiable claim issuer is able to rely on an authentication and validation performed by the identity provider.

In the embodiments disclosed herein, the digital wallet generates the identification value. The digital wallet then sends a request to the identity provider for an authentication token so that the digital wallet can receive one or more verifiable claims from the verifiable claim issuer. The identification value is included in the request for the authentication token.

Upon receipt of the request for the authentication token, the identity provider authenticates or validates the request is such a way that the identity provider is able to ascertain that the owner of the digital wallet is an entity that is trusted to receive services from the verifiable claim issuer. The identity provider then generates an authentication token and includes the identification value in the authentication token. However, the identity provider does not decrypt or otherwise interpret the identification value, but merely includes the identification value in the authentication token. The authentication token is then returned to the digital wallet.

The digital wallet generates a request for one or more verifiable claims and includes the authentication token and the DID in the request. The request is then sent to the verifiable claim issuer. Upon receipt of the request, the verifiable claim issuer authenticates or validates the authentication token to ascertain that the owner of the digital wallet is authorized to receive verifiable claims from the verifiable claim issuer. In addition, the verifiable claim issuer uses the DID to validate the identification value. Thus, the embodiments disclosed herein bind the authentication proofs generated by the identity provider to the DID of the DID owner requesting the verifiable claims. This provides a high level of security and allows the verifiable claim issuer to have a high level of trust that the party that requested the verifiable claims that will be bound to the DID is the actual subject of the verifiable claims.

In one embodiment, a digital wallet generates an identification value. The digital wallet generates a first request including the identification value for an authentication token from an identification provider. The first request is provided to the identification provider. The digital wallet receives from the identification provider, in response to the identification provider validating the first request, the authentication token that authenticates the digital wallet with a verifiable claim issuer including the identification value. The digital wallet generates a second request for one or more

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verifiable claims from the verifiable claim issuer. The second request includes the DID and authentication token including the identification value. In response to the verifiable claim issuer validating the authentication token and the identification value, one or more verifiable claims from the verifiable claim issuer are received by the digital wallet.

In an embodiment, the identification value is a random number or hash value that is signed using the private key. In an embodiment, the identification value is a hash of the DID. In an embodiment, the identification value is the DID. In an embodiment, the public key is accessible by accessing a DID document associated with the DID that is tied to the distributed ledger. In an embodiment, the first request includes identification information related to the DID owner such as a username, password, and information requesting the authentication token. In an embodiment, the authentication token includes one or more authentication proofs that are generated by the identity provider. In an embodiment, the authentication token is cryptographically signed by the identity provider.

In the embodiment, the second request includes information specifying one or more verifiable claims that the DID owner desires to receive from the verifiable claim issuer. In the embodiment, the second request is cryptographically signed using the same private key as was used to generate the identification value. In the embodiment, the one or more verifiable claims received from the verifiable claim issuer are stored at the digital wallet and/or in an identity hub.

In one embodiment, an identity provider receives an authentication request for an authentication token from a digital wallet that is associated with a DID of a DID owner. The authentication token authenticates the digital wallet with a verifiable claim issuer. The authentication request includes an identification value associated with the DID that was generated by the digital wallet. The authentication request is validated. In response to validating the authentication request, the identity provider generates the authentication token including the identification value and provides the authentication token to the digital wallet.

In an embodiment, the authentication request includes identification information related to the DID owner and information requesting the authentication token. In an embodiment, the authentication token includes one or more authentication proofs that are generated by the identity provider. In an embodiment, the identity provider cryptographically signs the authentication token. In an embodiment, the identity provider includes the identification value with the authentication token without decrypting the identification value.

In one embodiment, a verifiable claim issuer receives a request for one or more verifiable claims from a digital wallet that is associated with a DID of a DID owner. The request includes the DID and an authentication token generated by an identity provider. The authentication token includes an identification value that was generated by the digital wallet and that is bound to the DID. The verifiable claim issuer uses the DID to access a public key that is associated with the private key via a distributed ledger. The verifiable claim issuer validates the authentication token to ensure that the identity provider issued the authentication token. The verifiable claim issuer validates the identification value. In response to validating the authentication token and the identification value, the verifiable claim issuer provides one or more verifiable claims to the digital wallet.

In the embodiment, the identification value is validated using a public key that is associated with a private key associated with the DID. In an embodiment, the identifica-

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tion value is validated by comparing a hash of the DID received in the request with a hash of the DID. In an embodiment, the identification value is validated by comparing the DID received in the request with the DID.

Additional features and advantages will be set forth in the description which follows, and in part will be obvious from the description, or may be learned by the practice of the teachings herein. Features and advantages of the invention may be realized and obtained by means of the instruments and combinations particularly pointed out in the appended claims. Features of the present invention will become more fully apparent from the following description and appended claims or may be learned by the practice of the invention as set forth hereinafter.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to describe the manner in which the above-recited and other advantages and features can be obtained, a more particular description of the subject matter briefly described above will be rendered by reference to specific embodiments which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments and are not therefore to be considered to be limiting in scope, embodiments will be described and explained with additional specificity and details through the use of the accompanying drawings in which:

FIG. 1 illustrates an example computing system in which the principles described herein may be employed;

FIG. 2 illustrates an example environment for creating a decentralized identification or identifier (DID);

FIG. 3 illustrates an example environment for various DID management operations and services;

FIG. 4 illustrates an example decentralized personal storage or identity hub;

FIG. 5 illustrates an example computing environment, in which the principles described herein are implemented;

FIG. 6 illustrates an example authentication request;

FIG. 7 illustrates an example authentication token;

FIG. 8 illustrates an example verifiable claim request;

FIG. 9A illustrates an example claim;

FIG. 9B illustrates an example verifiable claim;

FIG. 10 illustrates a process flow of the computing environment of FIG. 5;

FIG. 11 illustrates a flow chart of an example method for a computing system of a digital wallet to receive an authentication token from an identity provider, the authentication token being used by the computing system in a request for one or more verifiable claims from a verifiable claim issuer so that the computing system can be validated by the verifiable claim issuer;

FIG. 12 illustrates a flow chart of an example method for a computing system of an identity provider to generate an authentication token for use by a digital wallet in a request for one or more verifiable claims from a verifiable claim issuer so that the digital wallet can be validated by the verifiable claim issuer; and

FIG. 13 illustrates a flow chart of an example method for a computing system of a verifiable claim issuer to provide one or more verifiable claims to a digital wallet, the computing system receiving an authentication token issued by an identity provider that validates that the digital wallet is authorized to receive the one or more verifiable claims.

DETAILED DESCRIPTION

In one embodiment, a digital wallet generates an identification associated with a DID of a DID owner. The digital

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wallet generates a first request including the identification value for an authentication token from an identification provider. The first request is provided to the identification provider. The digital wallet receives from the identification provider, in response to the identification provider validating the first request, the authentication token that authenticates the digital wallet with a verifiable claim issuer including the identification value. The digital wallet generates a second request for one or more verifiable claims from the verifiable claim issuer. The second request includes the DID and authentication token including the identification value. In response to the verifiable claim issuer validating the authentication token and the identification value, one or more verifiable claims from the verifiable claim issuer are received by the digital wallet.

In one embodiment, an identity provider receives an authentication request for an authentication token from a digital wallet that is associated with a DID of a DID owner. The authentication token authenticates the digital wallet with a verifiable claim issuer. The authentication request includes an identification value associated with the DID that was generated by the digital wallet. The authentication request is validated. In response to validating the authentication request, the identity provider generates the authentication token including the identification value and provides the authentication token to the digital wallet.

In one embodiment, a verifiable claim issuer receives a request for one or more verifiable claims from a digital wallet that is associated with a DID of a DID owner. The request includes the DID and an authentication token generated by an identity provider. The authentication token includes an identification value that was generated by the digital wallet and that is bound to the DID. The verifiable claim issuer validates the authentication token to ensure that the identity provider issued the authentication token. The verifiable claim issuer validates the identification value. In response to validating the authentication token and the identification value, the verifiable claim issuer provides one or more verifiable claims to the digital wallet.

Because the principles described herein are performed in the context of a computing system, some introductory discussion of a computing system will be described with respect to FIG. 1. Then, this description will return to the principles of the embodiments disclosed herein with respect to the remaining figures.

Computing systems are now increasingly taking a wide variety of forms. Computing systems may, for example, be handheld devices, appliances, laptop computers, desktop computers, mainframes, distributed computing systems, data centers, or even devices that have not conventionally been considered a computing system, such as wearables (e.g., glasses). In this description and in the claims, the term “computing system” is defined broadly as including any device or system (or a combination thereof) that includes at least one physical and tangible processor, and a physical and tangible memory capable of having thereon computer-executable instructions that are executed by a processor. The memory takes any form and depends on the nature and form of the computing system. A computing system is distributed over a network environment and includes multiple constituent computing systems.

As illustrated in FIG. 1, in its most basic configuration, a computing system 100 typically includes at least one hardware processing unit 102 and memory 104. The at least one hardware processing unit 102 includes a general-purpose processor and also includes a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC),

or any other specialized circuit. The memory **104** is physical system memory, which is volatile, non-volatile, or some combination of the two. The term “memory” also be used herein to refer to non-volatile mass storage such as physical storage media. If the computing system is distributed, the processing, memory and/or storage capability is distributed as well.

The computing system **100** also has thereon multiple structures often referred to as an “executable component”. For instance, memory **104** of the computing system **100** is illustrated as including executable component **106**. The term “executable component” is the name for a structure that is well understood to one of ordinary skill in the art in the field of computing as being a structure that can be software, hardware, or a combination thereof. For instance, when implemented in software, one of ordinary skill in the art would understand that the structure of an executable component include software objects, routines, methods, and so forth, that is executed on the computing system, whether such an executable component exists in the heap of a computing system, or whether the executable component exists on computer-readable storage media.

In such a case, one of ordinary skill in the art will recognize that the structure of the executable component exists on a computer-readable medium such that, when interpreted by one or more processors of a computing system (e.g., by a processor thread), the computing system is caused to perform a function. Such a structure is computer-readable directly by the processors (as is the case if the executable component were binary). Alternatively, the structure is structured to be interpretable and/or compiled (whether in a single stage or in multiple stages) so as to generate such binary that is directly interpretable by the processors. Such an understanding of example structures of an executable component is well within the understanding of one of ordinary skill in the art of computing when using the term “executable component”.

The term “executable component” is also well understood by one of ordinary skill as including structures, such as hardcoded or hard-wired logic gates, that are implemented exclusively or near-exclusively in hardware, such as within a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or any other specialized circuit. Accordingly, the term “executable component” is a term for a structure that is well understood by those of ordinary skill in the art of computing, whether implemented in software, hardware, or a combination. In this description, the terms “component”, “agent”, “manager”, “service”, “engine”, “module”, “virtual machine” or the like also be used. As used in this description and in the case, these terms (whether expressed with or without a modifying clause) are also intended to be synonymous with the term “executable component”, and thus also have a structure that is well understood by those of ordinary skill in the art of computing.

In the description that follows, embodiments are described with reference to acts that are performed by one or more computing systems. If such acts are implemented in software, one or more processors (of the associated computing system that performs the act) direct the operation of the computing system in response to having executed computer-executable instructions that constitute an executable component. For example, such computer-executable instructions are embodied on one or more computer-readable media that form a computer program product. An example of such an operation involves the manipulation of data. If such acts are implemented exclusively or near-exclusively in hardware, such as within an FPGA or an ASIC, the computer-

executable instructions are hardcoded or hard-wired logic gates. The computer-executable instructions (and the manipulated data) are stored in the memory **104** of the computing system **100**. Computing system **100** also contain communication channels **108** that allow the computing system **100** to communicate with other computing systems over, for example, network **110**.

While not all computing systems require a user interface, in some embodiments, the computing system **100** includes a user interface system **112** for use in interfacing with a user. The user interface system **112** includes output mechanisms **112A** as well as input mechanisms **112B**. The principles described herein are not limited to the precise output mechanisms **112A** or input mechanisms **112B** as such will depend on the nature of the device. However, output mechanisms **112A** might include, for instance, speakers, displays, tactile output, holograms and so forth. Examples of input mechanisms **112B** might include, for instance, microphones, touchscreens, holograms, cameras, keyboards, mouse or other pointer input, sensors of any type, and so forth.

Embodiments described herein comprise or utilize a special purpose or general-purpose computing system including computer hardware, such as, for example, one or more processors and system memory, as discussed in greater detail below. Embodiments described herein also include physical and other computer-readable media for carrying or storing computer-executable instructions and/or data structures. Such computer-readable media can be any available media that can be accessed by a general-purpose or special-purpose computing system. Computer-readable media that store computer-executable instructions are physical storage media. Computer-readable media that carry computer-executable instructions are transmission media. Thus, by way of example, and not limitation, embodiments of the invention can comprise at least two distinctly different kinds of computer-readable media: storage media and transmission media.

Computer-readable storage media includes RAM, ROM, EEPROM, CD-ROM, or other optical disk storage, magnetic disk storage, or other magnetic storage devices, or any other physical and tangible storage medium which can be used to store desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general-purpose or special-purpose computing system.

A “network” is defined as one or more data links that enable the transport of electronic data between computing systems and/or modules and/or other electronic devices. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a computing system, the computing system properly views the connection as a transmission medium. Transmissions media can include a network and/or data links which can be used to carry desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general-purpose or special-purpose computing system. Combinations of the above should also be included within the scope of computer-readable media.

Further, upon reaching various computing system components, program code means in the form of computer-executable instructions or data structures can be transferred automatically from transmission media to storage media (or vice versa). For example, computer-executable instructions or data structures received over a network or data link can be buffered in RAM within a network interface module (e.g., a “NIC”), and then eventually transferred to computing

system RAM and/or to less volatile storage media at a computing system. Thus, it should be understood that storage media can be included in computing system components that also (or even primarily) utilize transmission media.

Computer-executable instructions comprise, for example, instructions and data which, when executed at a processor, cause a general-purpose computing system, special purpose computing system, or special purpose processing device to perform a certain function or group of functions. Alternatively, or in addition, the computer-executable instructions configure the computing system to perform a certain function or group of functions. The computer executable instructions are, for example, binaries or even instructions that undergo some translation (such as compilation) before direct execution by the processors, such as intermediate format instructions such as assembly language, or even source code.

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the described features or acts described above. Rather, the described features and acts are disclosed as example forms of implementing the claims.

Those skilled in the art will appreciate that the invention is practiced in network computing environments with many types of computing system configurations, including, personal computers, desktop computers, laptop computers, message processors, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, mobile telephones, PDAs, pagers, routers, switches, data centers, wearables (such as glasses) and the like. In some cases, the invention also is practiced in distributed system environments where local and remote computing systems, which are linked (either by hardwired data links, wireless data links, or by a combination of hardwired and wireless data links) through a network, both perform tasks. In a distributed system environment, program modules are located in both local and remote memory storage devices.

Those skilled in the art will also appreciate that the invention is practiced in a cloud computing environment. Cloud computing environments are distributed, although this is not required. When distributed, cloud computing environments are distributed internationally within an organization and/or have components possessed across multiple organizations. In this description and the following claims, “cloud computing” is defined as a model for enabling on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services). The definition of “cloud computing” is not limited to any of the other numerous advantages that can be obtained from such a model when properly deployed.

The remaining figures discuss various computing system which corresponds to the computing system **100** previously described. The computing systems of the remaining figures include various components or functional blocks that implement the various embodiments disclosed herein as will be explained. The various components or functional blocks are implemented on a local computing system or are implemented on a distributed computing system that includes elements resident in the cloud or that implement aspects of cloud computing. The various components or functional blocks are implemented as software, hardware, or a combination of software and hardware. The computing systems of the remaining figures include more or less than the components illustrated in the figures and some of the components are combined as circumstances warrant. Although not nec-

essarily illustrated, the various components of the computing systems access and/or utilize a processor and memory, such as the hardware processing unit **102** and memory **104**, as needed to perform their various functions.

Some introductory discussions of a decentralized identification (DID) and the environment in which they are created and reside will now be given with respect to FIG. 2, which illustrates a decentralized network **200**. As illustrated in FIG. 2, a DID owner **201** owns or controls a DID **205** that represents an identity of the DID owner **201**. The DID owner **201** registers a DID using a creation and registration service, which will be explained in more detail below.

The DID owner **201** is any entity that could benefit from a DID. For example, the DID owner **201** is a human being or an organization of human beings. Such organizations might include a company, department, government, agency, or any other organization or group of organizations. Each individual human being might have a DID while the organization(s) to which each belongs might likewise have a DID.

The DID owner **201** alternatively may be a machine, system, or device, or a collection of machine(s), device(s) and/or system(s). In still other embodiments, the DID owner **201** is a subpart of a machine, system, or device. For instance, a device could be a printed circuit board, where the subpart of that circuit board are individual components of the circuit board. In such embodiments, the machine or device has a DID and each subpart also have a DID. A DID owner might also be a software component such as the executable component **106** described above with respect to FIG. 1. An example of a complex executable component **106** might be an artificial intelligence. An artificial intelligence also owns a DID.

Thus, the DID owner **201** is any reasonable entity, human or non-human, that is capable of creating the DID **205** or at least having the DID **205** created for and associated with them. Although the DID owner **201** is shown as having a single DID **205**, this need not be the case as there may be any number of DIDs associated with the DID owner **201** as circumstances warrant.

As mentioned, the DID owner **201** creates and registers the DID **205**. The DID **205** is any identifier that is associated with the DID owner **201**. Preferably, that identifier is unique to that DID owner **201**, at least within a scope in which the DID is anticipated to be in use. As an example, the identifier is a locally unique identifier, and perhaps more desirably a globally unique identifier for identity systems anticipated to operate globally. In some embodiments, the DID **205** is a Uniform Resource Identifier (URI) (such as a Uniform Resource Locator (URL)) or other pointers that relates the DID owner **201** to mechanism to engage in trustable interactions with the DID owner **201**.

The DID **205** is “decentralized” because it does not require a centralized, third-party management system for generation, management, or use. Accordingly, the DID **205** remains under the control of the DID owner **201**. This is different from conventional centralized IDs based trust on centralized authorities and that remain under control of the corporate directory services, certificate authorities, domain name registries, or other centralized authority (referred to collectively as “centralized authorities” herein). Accordingly, the DID **205** is any identifier that is under the control of the DID owner **201** and independent of any centralized authority.

In some embodiments, the structure of the DID **205** is as simple as a username or some other human-understandable term. However, in other embodiments, the DID **205** prefer-

ably be a random string of numbers and letters for increased security. In one embodiment, the DID **205** is a string of **128** letters and numbers. Accordingly, the embodiments disclosed herein are not dependent on any specific implementation of the DID **205**. In a very simple example, the DID **205** is shown as “123 ABC”.

As also shown in FIG. 2, the DID owner **201** has control of a private key **206** and public key **207** pair that are associated with the DID **205**. Because the DID **205** is independent of any centralized authority, the private key **206** should at all times be fully in control of the DID owner **201**. That is, the private and public keys should be generated in a decentralized manner that ensures that they remain under the control of the DID owner **201**.

As will be described in more detail to follow, the private key **206** and public key **207** pair is generated on a device controlled by the DID owner **201**. The private key **206** and public key **207** pairs should not be generated on a server controlled by any centralized authority as this causes the private key **206** and public key **207** pairs to not be fully under the control of the DID owner **201** at all times. Although FIG. 2 and this description have described a private and public key pair, it will also be noted that other types of reasonable cryptographic information and/or mechanism also be used as circumstances warrant.

FIG. 2 also illustrates a DID document **210** that is associated with the DID **205**. As will be explained in more detail to follow, the DID document **210** is generated at the time that the DID **205** is created. In its simplest form, the DID document **210** describes how to use the DID **205**. Accordingly, the DID document **210** includes a reference to the DID **205**, which is the DID that is described by the DID document **210**. In some embodiments, the DID document **210** is implemented according to methods specified by a distributed ledger **220** that will be used to store a representation of the DID **205** as will be explained in more detail to follow. Thus, the DID document **210** has different methods depending on the specific distributed ledger.

The DID document **210** also includes the public key **207** created by the DID owner **201** or some other equivalent cryptographic information. The public key **207** is used by third-party entities that are given permission by the DID owner **201** to access information and data owned by the DID owner **201**. The public key **207** may also be used by verifying that the DID owner **201**, in fact, owns or controls the DID **205**.

The DID document **210** also includes authentication information **211**. The authentication information **211** specify one or more mechanisms by which the DID owner **201** is able to prove that the DID owner **201** owns the DID **205**. In other words, the mechanisms of authentication information **211** show proof of a binding between the DID **205** (and thus the DID owner **201**) and the DID document **210**. In one embodiment, the authentication information **211** specifies that the public key **207** be used in a signature operation to prove the ownership of the DID **205**. Alternatively, or in addition, the authentication information **211** specifies that the public key **207** be used in a biometric operation to prove ownership of the DID **205**. Accordingly, the authentication information **211** includes any number of mechanisms by which the DID owner **201** is able to prove that the DID owner **201** owns the DID **205**.

The DID document **210** also includes authorization information **212**. The authorization information **212** allows the DID owner **201** to authorize third party entities the rights to modify the DID document **210** or some part of the document without giving the third party the right to prove ownership

of the DID **205**. For example, the authorization information **212** allows the third party to update any designated set of one or more fields in the DID document **210** using any designated update mechanism. Alternatively, the authorization information allows the third party to limit the usages of DID **205** by the DID owner **201** for a specified time period. This is useful when the DID owner **201** is a minor child and the third party is a parent or guardian of the child. The authorization information **212** allows the parent or guardian to limit the use of the DID **205** until such time as the child is no longer a minor.

The authorization information **212** also specifies one or more mechanisms that the third party will need to follow to prove they are authorized to modify the DID document **210**. In some embodiments, this mechanism is similar to those discussed previously with respect to the authentication information **211**.

The DID document **210** also includes one or more service endpoints **213**. A service endpoint includes a network address at which a service operates on behalf of the DID owner **201**. Examples of specific services include discovery services, social networks, file storage services such as identity servers or hubs, and verifiable claim repository services. Accordingly, the service endpoints **213** operate as pointers for the services that operate on behalf of the DID owner **201**. These pointers are used by the DID owner **201** or by third party entities to access the services that operate on behalf of the DID owner **201**. Specific examples of service endpoints **213** will be explained in more detail to follow.

The DID document **210** further includes identification information **214**. The identification information **214** includes personally identifiable information such as the name, address, occupation, family members, age, hobbies, interests, or the like of DID owner **201**. Accordingly, the identification information **214** listed in the DID document **210** represents a different persona of the DID owner **201** for different purposes. For instance, a persona is pseudo-anonymous, e.g., the DID owner **201** include a pen name in the DID document when identifying him or her as a writer posting articles on a blog; a persona is fully anonymous, e.g., the DID owner **201** only want to disclose his or her job title or other background data (e.g., a school teacher, an FBI agent, an adult older than 21 years old, etc.) but not his or her name in the DID document; and a persona is specific to who the DID owner **201** is as an individual, e.g., the DID owner **201** includes information identifying him or her as a volunteer for a particular charity organization, an employee of a particular corporation, an award winner of a particular award, etc.

The DID document **210** also includes credential information **215**, which may also be referred to herein as an attestation. The credential information **215** (also referred to as a verifiable claim) is any information that is associated with the DID owner **201**'s background. For instance, the credential information **215** is (but not limited to) a qualification, an achievement, a government ID, a government right such as a passport or a driver's license, a digital asset provider or bank account, a university degree or other educational history, employment status and history, or any other information about the DID owner **201**'s background.

The DID document **210** also includes various other information **216**. In some embodiments, the other information **216** includes metadata specifying when the DID document **210** was created and/or when it was last modified. In other embodiments, the other information **216** includes cryptographic proofs of the integrity of the DID document **210**. In still further embodiments, the other information **216**

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includes additional information that is either specified by the specific method implementing the DID document or desired by the DID owner 201.

FIG. 2 also illustrates a distributed ledger or blockchain 220. The distributed ledger 220 is any decentralized, distributed network that includes various computing systems that are in communication with each other. For example, the distributed ledger 220 includes a first distributed computing system 230, a second distributed computing system 240, a third distributed computing system 250, and any number of additional distributed computing systems as illustrated by the ellipses 260. The distributed ledger or blockchain 220 operates according to any known standards or methods for distributed ledgers. Examples of conventional distributed ledgers that correspond to the distributed ledger or blockchain 220 include, but are not limited to, Bitcoin [BTC], Ethereum, and Litecoin.

In the context of DID 205, the distributed ledger or blockchain 220 is used to store a representation of the DID 205 that points to the DID document 210. In some embodiments, the DID document 210 is stored on the actual distributed ledger itself. Alternatively, in other embodiments the DID document 210 is stored in a data storage (not illustrated) that is associated with the distributed ledger or blockchain 220.

As mentioned, a representation of the DID 205 is stored on each distributed computing system of the distributed ledger or blockchain 220. For example, in FIG. 2 this is shown as the DID has 231, DID has 241, and DID has 251, which are ideally identical copies of the same DID. The DID hash 231, DID hash 241, and DID hash 251 then point to the location of the DID document 210. The distributed ledger or blockchain 220 also store numerous other representations of other DIDs as illustrated by references 232, 233, 234, 242, 243, 244, 252, 253, and 254.

In one embodiment, when the DID owner 201 creates the DID 205 and the associated DID document 210, the DID hash 231, DID hash 241, and DID hash 251 are written to the distributed ledger or blockchain 220. The distributed ledger or blockchain 220 thus records that the DID 205 now exists. Since the distributed ledger or blockchain 220 is decentralized, the DID 205 is not under the control of any entity outside of the DID owner 201. The DID hash 231, DID hash 241, and DID hash 251 includes, in addition to the pointer to the DID document 210, a record or timestamp that specifies when the DID 205 was created. At a later date when modifications are made to the DID document 210, this also is recorded in DID hash 231, DID hash 241, and DID hash 251. The DID hash 231, DID hash 241, and DID hash 251 further includes a copy of the public key 207 so that the DID 205 is cryptographically bound to the DID document 210.

Having described DIDs and how they operate generally with reference to FIG. 2, specific embodiments of DID environments will now be explained. Turning to FIG. 3, a computing system environment 300 that is used to perform various DID management operations and services will now be explained. It will be appreciated that the environment of FIG. 3 reference elements from FIG. 2 as needed for ease of explanation.

As shown in FIG. 3, the computing system environment 300 includes various devices and computing systems that are owned or otherwise under the control of the DID owner 201. These include a user device 301. The user device 301 is, but is not limited to, a mobile device such as a smart phone, a computing device such as a laptop computer, or any device such as a car or an appliance that includes computing abilities. The user device 301 includes a web browser 302

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operating on the device and an operating system 303 operating the device. More broadly speaking, the dashed line 304 represents that all of these devices are owned or otherwise under the control of the DID owner 201.

The computing system environment 300 also includes a DID management module 320. It will be noted that in operation, the DID management module 320 resides on and is executed by one or more of user device 301, web browser 302, and the operating system 303 as illustrated by respective lines 301a, 302a, and 303a. Accordingly, the DID management module 320 is shown as being separate for ease of explanation. In some embodiments, the DID management module 320 is referred to as a “digital wallet”, a “wallet app”, or a “user agent”. It will be appreciated by one of skill in the art, however, that a digital wallet, wallet app, or user agent can be implemented in a computing system other than the DID management module 320 in other embodiments.

As shown in FIG. 3, the DID management module 320 includes a DID creation module 330. The DID creation module 330 is used by the DID owner 201 to create the DID 205 or any number of additional DIDs, such as DID 331. In one embodiment, the DID creation module includes or otherwise has access to a User Interface (UI) element 335 that guides the DID owner 201 in creating the DID 205. The DID creation module 330 has one or more drivers that are configured to work with specific distributed ledgers such as distributed ledger 220 so that the DID 205 complies with the underlying methods of that distributed ledger.

A specific embodiment will now be described. For example, the UI 335 prompt for the user to enter a username or some other human recognizable name. This name is used as a display name for the DID 205 that will be generated. As previously described, the DID 205 is a long string of random numbers and letters and so having a human-recognizable name for a display name is advantageous. The DID creation module 330 then generates the DID 205. In the embodiments having the UI 335, the DID 205 is shown in a listing of identities and is associated with the human-recognizable name.

The DID creation module 330 also included a key generation module 350. The key generation module generates the private key 206 and public key 207 pair previously described. The DID creation module 330 uses the DID 205 and the private and public key pair to generate the DID document 210. Thus, the DID 205 is signed by the DID owner 201 using the private key 206 when generating the DID and the DID document.

In operation, the DID creation module 330 accesses a registrar 310 that is configured to the specific distributed ledger that will be recording the transactions related to the DID 205. The DID creation module 330 uses the registrar 310 to record DID hash 231, DID hash 241, and DID hash 251 in the distributed ledger in the manner previously described, and to store the DID document 210 in the manner previously described. This process uses the public key 207 in the hash generation.

In some embodiments, the DID management module 320 includes an ownership module 340. The ownership module 340 provides mechanisms that ensure that the DID owner 201 is in sole control of the DID 205. In this way, the provider of the DID management module 320 is able to ensure that the provider does not control the DID 205 but is only providing the management services.

As previously discussed, the key generation module 350 generates the private key 206 and public key 207 pair and the public key 207 is then recorded in the DID document 210. Accordingly, the public key 207 is usable by all devices

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associated with the DID owner **201** and all third parties that desire to provide services to the DID owner **201**. Accordingly, when the DID owner **201** desires to associate a new device with the DID **205**, the DID owner **201** executes the DID creation module **330** on the new device. The DID creation module **330** then uses the registrar **310** to update the DID document **210** to reflect that the new device is now associated with the DID **205**, which update would be reflected in a transaction on the distributed ledger **220**, as previously described.

In some embodiments, however, it is advantageous to have a public key per user device **301** owned by the DID owner **201** as this allows the DID owner **201** to sign with the device-specific public key without having to access a general public key. In other words, since the DID owner **201** will use different devices at different times (for example using a mobile phone in one instance and then using a laptop computer in another instance), it is advantageous to have a key associated with each device to provide efficiencies in signing using the keys. Accordingly, in such embodiments the key generation module **350** generates additional public keys **208** and **209** when the additional devices execute the DID creation module **330**. These additional public keys are associated with the private key **206** or in some instances are paired with a new private key.

In those embodiments where the additional public keys **208** and **209** are associated with different devices, the additional public keys **208** and **209** are recorded in the DID document **210** as being associated with those devices. This is shown in FIG. 3. It will be appreciated that the DID document **210** often includes the information (information **205**, **207** and **211** through **216**) previously described in relation to FIG. 2 in addition to the information (information **208**, **209** and **365**) shown in FIG. 3. If the DID document **210** existed prior to the device-specific public keys being generated, then the DID document **210** would be updated by the DID creation module **330** via the registrar **310** and this would be reflected in an updated transaction on the distributed ledger **220**.

In some embodiments, the DID owner **201** often desires to keep secret the association of a device with a public key or the association of a device with the DID **205**. Accordingly, the DID creation module **330** causes that such data be secretly shown in the DID document **210**.

As described thus far, the DID **205** has been associated with all the devices under the control of the DID owner **201**, even when the devices have their own public keys. However, in some embodiments it may be useful for each device or some subset of devices under the control of the DID owner **201** to each have their own DID. Thus, in some embodiments the DID creation module **330** generates an additional DID, for example DID **331**, for each device. The DID creation module **330** then generates private and public key pairs and DID documents for each of the devices and has them recorded on the distributed ledger **220** in the manner previously described. Such embodiments are advantageous for devices that change ownership as it is possible to associate the device specific DID to the new owner of the device by granting the new owner authorization rights in the DID document and revoking such rights from the old owner.

As mentioned, to ensure that the private key **206** is totally in the control of the DID owner **201**, the private key **206** is created on the user device **301**, web browser **302**, or operating system **303** that is owned or controlled by the DID owner **201** that executed the DID management module **320**. In this way, there is little chance that of a third-party (and

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most consequentially, the provider of the DID management module **320**) gaining control of the private key **206**.

However, there is a chance that the device storing the private key **206** is lost by the DID owner **201**, which causes the DID owner **201** to lose access to the DID **205**. Accordingly, in some embodiments, the UI **335** includes the option to allow the DID owner **201** to export the private key **206** to an off device secured database **305** that is under the control of the DID owner **201**. As an example, the database **305** is one of the identity hubs **410** described below with respect to FIG. 4. A storage module **380** is configured to store data (such as the private key **206** or the credential information **215** made by or about the DID owner **201**) off device in the database **305** or in the identity hubs **410** that will be described in more detail to follow. Of course, in some embodiments the storage module **380** stores at least some data on the device if the device has sufficient storage resources. In some embodiments, the private key **206** is stored as a QR code that is scanned by the DID owner **201**.

In other embodiments, the DID management module **320** includes a recovery module **360** that is used to recover a lost private key **206**. In operation, the recovery module **360** allows the DID owner **201** to select one or more recovery mechanisms **365** at the time the DID **205** is created that are later used to recover the lost private key. In those embodiments having the UI **335**, the UI **335** allows the DID owner **201** to provide information that will be used by the one or more recovery mechanisms **365** during recovery. The recovery module **360** run on any device associated with the DID **205**.

The DID management module **320** also included a revocation module **370** that is used to revoke or sever a device from the DID **205**. In operation, the revocation module uses the UI **335**, which allows the DID owner **201** to indicate a desire to remove a device from being associated with the DID **205**. In one embodiment, the revocation module **370** accesses the DID document **210** and causes all references to the device to be removed from the DID document **210**. Alternatively, the public key for the device is removed. This change in the DID document **210** is then reflected as an updated transaction on the distributed ledger **220** as previously described.

FIG. 4 illustrates an embodiment of a computing system environment **400** in which a DID such as DID **205** is utilized. Specifically, the computing system environment **400** is used to describe the use of the DID **205** in relation to one or more decentralized stores or identity hubs **410** that are each under the control of the DID owner **201** to store data belonging to or regarding the DID owner **201**. For instance, data is stored within the identity hubs using the storage module **380** of FIG. 3. It will be noted that FIG. 4 includes references to elements first discussed in relation to FIG. 2 or 3 and thus uses the same reference numeral for ease of explanation.

In one embodiment, the identity hubs **410** are multiple instances of the same identity hub. This is represented by the line **410A**. Thus, the various identity hubs **410** include at least some of the same data and services. Accordingly, if a change is made to part of at least some of the data (and potentially any part of any of the data) in one of the identity hubs **410**, the change is reflected in one or more of (and perhaps all of) the remaining identity hubs.

The identity hubs **410** may be any data store that is in the exclusive control of the DID owner **201**. As an example only, the first identity hub **411** and second identity hub **412** are implemented in cloud storage (perhaps within the same cloud, or even on different clouds managed by different

cloud providers) and thus are able to hold a large amount of data. Accordingly, a full set of the data is storable in these identity hubs.

However, the identity hubs **413** and **414** may have less memory space. Accordingly, in these identity hubs a descriptor of the data stored in the first and second identity hubs is included. Alternatively, a record of changes made to the data in other identity hubs is included. Thus, changes in one of the identity hubs **410** are either fully replicated in the other identity hubs or at least a record or descriptor of that data is recorded in the other identity hubs.

Because the identity hubs are multiple instances of the same identity hub, only a full description of the first identity hub **411** will be provided as this description also applies to the identity hubs **412** through **414**. As illustrated, identity hub **411** includes data storage **420**. The data storage **420** is used to store any type of data that is associated with the DID owner **201**. In one embodiment the data is a collection **422** of a specific type of data corresponding to a specific protocol. For example, the collection **422** may be medical records data that corresponds to a specific protocol for medical data. The collection **422** also includes other types of data, such as credential information **215** made by or about the DID owner **201**.

In one embodiment, the stored data has different authentication and privacy settings **421** associated with the stored data. For example, a first subset of the data has a setting **421** that allows the data to be publicly exposed, but that does not include any authentication to the DID owner **201**. This type of data is typically for relatively unimportant data such as color schemes and the like. A second subset of the data has a setting **421** that allows the data to be publicly exposed and that includes authentication to the DID owner **201**. A third subset of the data has a setting **421** that encrypts the subset of data with the private key **206** and public key **207** pair (or some other key pair) associated with the DID owner **201**. This type of data will require a party to have access to the public key **207** (or to some other associated public key) in order to decrypt the data. This process also includes authentication to the DID owner **201**. A fourth subset of the data has a setting **421** that restricts this data to a subset of third parties. This requires that public keys associated with the subset of third parties be used to decrypt the data. For example, the DID owner **201** causes the setting **421** to specify that only public keys associated with friends of the DID owner **201** are able to decrypt this data. With respect to data stored by the storage module **380**, these settings **421** are at least partially composed by the storage module **380** of FIG. 3.

In some embodiments, the identity hub **411** has a permissions module **430** that allows the DID owner **201** to set specific authorization or permissions for third parties such as third parties **401** and **402** to access the identity hub. For example, the DID owner **201** provides access permission to his or her spouse to all the data stored in data storage **420**. Alternatively, the DID owner **201** allows access to his or her doctor for any medical records. It will be appreciated that the DID owner **201** is able to give permission to any number of third parties to access a subset of the data stored in data storage **420**. This will be explained in more detail to follow. With respect to data stored by the storage module **380**, these access permissions **430** are at least partially composed by the storage module **380** of FIG. 3.

The identity hub **411** also include a messaging module **440**. In operation, the messaging module allows the identity hub to receive messages such as requests from parties such as third parties **401** and **402** to access the data and services

of the identity hub. In addition, the messaging module **440** allows the identity hub **411** to respond to the messages from the third parties and to also communicate with a DID resolver **450**. This will be explained in more detail to follow. The ellipsis **416** represents that the identity hub **411** may have additional services as circumstances warrant.

In one embodiment, the DID owner **201** wishes to authenticate a new user device **301** with the identity hub **411** that is already associated with the DID **205** in the manner previously described. Accordingly, the DID owner **201** utilizes the DID management module **320** associated with the new user device **301** to send a message to the identity hub **411** asserting that the new user device is associated with the DID **205** of the DID owner **201**.

However, the identity hub **411** is not able to initially recognize the new device as being owned by the DID owner **201**. Accordingly, the identity hub **411** uses the messaging module **440** to contact the DID resolver **450**. The message sent to the DID resolver **450** includes the DID **205**.

The DID resolver **450** is a service, application, or module that is configured in operation to search the distributed ledger **220** for DID documents associated with DIDs. Accordingly, in the embodiment the DID resolver **450** searches the distributed ledger **220** using the DID **205**, which should result in the DID resolver **450** finding the DID document **210**. The DID document **210** is then provided to the identity hub **411**.

As discussed previously, the DID document **210** includes a public key **208** or **209** that is associated with the new user device **301**. To verify that the new user device is owned by the DID owner **201**, the identity hub **411** provides a cryptographic challenge to the new user device **301** using the messaging module **440**. This cryptographic challenge is structured such that only a device having access to the private key **206** will be able to successfully answer the challenge.

In this embodiment, since the new user device is owned by DID owner **201** and thus has access to the private key **206**, the challenge is successfully answered. The identity hub **411** then records in the permissions **430** that the new user device **301** is able to access the data and services of the identity hub **411** and also the rest of the identity hubs **410**.

It will be noted that this process of authenticating the new user device **301** was performed without the need for the DID owner **201** to provide any username, password, or the like to the provider of the identity hub **411** (i.e., the first cloud storage provider) before the identity hub **411** could be accessed. Rather, the access was determined in a decentralized manner based on the DID **205**, the DID document **210**, and the associated public and private keys. Since these were at all times in the control of the DID owner **201**, the provider of the identity hub **411** was not involved and thus has no knowledge of the transaction or of any personal information of the DID owner **201**.

In another example embodiment, the DID owner **201** provides the DID **205** to the third-party **401** so that the third-party is able to access data or services stored on the identity hub **411**. For example, the DID owner **201** is a human who is at a scientific conference who desires to allow the third-party **401**, who is also a human, access to his or her research data. Accordingly, the DID owner **201** provides the DID **205** to the third-party **401**.

Once the third-party **401** has access to the DID **205**, he or she accesses the DID resolver **450** to access the DID document **210**. As previously discussed, the DID document **210** includes a service end point **213** that is an address or pointer to services associated with the decentralized identity.

Completing the research data example, the third-party **401** sends a message to the messaging module **440** asking for permission to access the research data. The messaging module **440** sends a message to the DID owner **201** asking if the third-party **401** should be given access to the research data. Because the DID owner desires to provide access to this data, the DID owner **201** allows permission to the third-party **401** and this permission is recorded in the permissions **430**.

The messaging module **440** then messages the third-party **401** informing the third-party that he or she is able to access the research data. The identity hub **411** and the third-party **401** directly communicate so that the third-party is able to access the data. It will be noted that in many cases, it will actually be an identity hub associated with the third-party **401** that communicates with the identity hub **411**. However, it may be a device of the third-party **401** that does the communication.

Advantageously, the above described process allows the identity hub **411** and the third-party **401** to communicate and to share the data without the need for the third-party to access the identity hub **411** in the conventional manner. Rather, the communication is provisioned in the decentralized manner using the DID **205** and the DID document **210**. This advantageously allows the DID owner to be in full control of the process.

As shown in FIG. 4, the third-party **402** also requests permission for access to the identity hub **411** using the DID **205** and the DID document **210**. Accordingly, the embodiments disclosed herein allow access to any number of third parties to the identity hubs **410**.

As briefly discussed above, the identity hub **411** is hosted in a cloud service. The service provider has access to the data stored in each user's identity hub **411**. Furthermore, the service provider also has access to certain activities of the DID owner. For example, the entities with whom the DID owner has shared his/her data is stored in the identity hub **411**. As another example, a user has multiple DIDs and has shared data amongst the multiple DIDs, alternatively, the user has used different DID management modules to access the same data. Based on the data sharing activities, the service provider of the identity hub **411** correlate the relationships of different DIDs and find out that two DIDs is related or owned by the same owner. Thus, the user's privacy is compromised.

The principles described herein will solve these potential privacy concerns of DID owners by encrypting the personal data stored in the identity hub **411**. The encryption/decryption keys are not stored or accessible by the identity hub **411**, so that the DID owners not only have great control over their data from other DID owners or users, but also have their privacy protected from the service providers.

There are many different objects stored in the identity hub **411**. A data object is a file, a folder, or any portion of data stored in the identity hub **411**. The whole identity hub **411** is encrypted with one encryption/decryption key as one object. Alternatively, a different portion of the data stored in the identity hub **411** is encrypted with different encryption/decryption keys.

In another example embodiment, verifiable claims (e.g., credential information **215**) are issued and stored at the identity hub **411**. For example, a verifiable claim that is associated with a DID owner **201** is issued by a claim issuing entity, and the issued verifiable claim is stored at the identity hub **411** that is associated with the DID owner **201**. The DID owner **201** send the verifiable claim to another entity when the other entity requires to verify the credential of the DID

owner. For example, the DID owner **201** is a person holding a driver's license, and the claim issuing entity is a DMV that has issued the DID owner's driver's license. The DMV issues a verifiable claim that verifies that the DID owner **201** is holding a valid driver's license. The DID owner **201** stores the verifiable claim in the identity hub **411**. Another entity is a rental car company, which requires the DID owner **201** to show that he/she has a valid driver's license. The DID owner then sends the verifiable claim stored at the identity hub **411** to the rental car company.

Having described DIDs and how they operate generally with reference to FIGS. 2-4, a specific embodiment of securing authentication flows using DIDs will now be explained with reference to FIG. 5. As illustrated in FIG. 5, a computing system environment **500** includes a digital wallet **510** that corresponds to the DID management module of FIG. 3. Thus, the digital wallet **510** is associated with the DID **205** and its associated private key **206** of the DID owner **201**. The DID **205** and private key **206** may be generated in the manner previously discussed.

The computing system environment **500** also includes a Verifiable Claim (VC) issuer **530**. The VC issuer **530** is an entity that is able to provide one or more verifiable claims **535** on behalf of the DID owner **201** who owns the digital wallet **510** and the DID **205**. The VC issuer **530** may be any reasonable entity that provides information that is associated with the DID owner **201** such as the information discussed previously in relation to credential information **215**.

The computing system environment **500** further includes an identification (ID) provider **520**. In operation, the ID provider **520** manages authentication and/or identity services on behalf of the VC issuer **530** so that the VC issuer will know that a request for one or more verifiable claims **535** comes from the DID owner **201** of the digital wallet **510** and the DID **205**. In other words, the ID provider **520** is a trusted source from the perspective of the VC issuer **530** and thus is able to provide a trusted verification of the identity (or at least the portion of the identity) of the DID owner **201** of the digital wallet **510** to the VC issuer **530**. The ID provider **520** may implement one or more of IndieAuth protocol, OAuth 2.0 protocol, OpenID Connect protocol, or any other reasonable authentication protocol.

The DID owner **201** of the digital wallet **510** may desire to receive one or more verifiable claims **535** that can be associated with the DID **205** from the issuer **530** as will be described in more detail to follow. The digital wallet **510** may therefore send a request to the VC issuer **530** along with the DID **205** to request the one or more verifiable claims **535**. The request will often include one or more authentication proofs generated by the ID **520** that verify the identity of the DID owner **201** of the digital wallet **510** to the VC issuer **530**. However, the VC issuer **530** may have no way to determine if the authentication proofs included in the request for the verifiable claims **535** are actually for the DID **205** or if some other DID being used in a malicious manner. Accordingly, the embodiments disclosed herein provide a mechanism for the digital wallet **510** to provide authentication proofs generated by the ID provider **520** to the VC issuer **530** and to bind the authentication proofs to the DID **205** as will now be explained.

The digital wallet **510** also includes an ID value generator **511**. In operation, the ID value generator **511** is configured to generate an ID value **512**. In some embodiments, the ID value **512** is a randomized string or hash value. In one embodiment, the ID value generator **511** uses the private key **206** to generate the ID value **512**. That is, the ID value generator **511** signs the ID value **512** using the private key

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206. In another embodiment, the ID generator **511** is configured to generate a hash of the DID **205** as the ID value **512**. In a further embodiment, the ID generator is configured to use the DID **205** as the generated ID value **512**. In each of these embodiments, the generation of the ID value **512** associates the ID value with the DID **205**, thus binding the ID value **512** to the DID **205**.

The digital wallet **510** further includes an authentication requestor **513**. In operation, the authentication requestor is configured to generate an authentication request **514**. In one embodiment, the authentication request **514** is a JSON document that conforms to the protocol implemented by the ID provider **520**. Of course, it will be appreciated that the authentication request **514** may be other than a JSON document as circumstances warrant. In the current embodiment, the ID value **512** is included in the authentication request **514**.

FIG. 6 illustrates a specific embodiment of the authentication request **514**. As illustrated, the authentication request **514** includes identification information **610**. The identification information **610** may be any information that is needed or required by the ID provider **520** so that the ID provider is able to authenticate that the authentication request **600** is from a known or trusted source. For example, the identification information **610** may include a username **611** and a password **622** that can be used by the ID provider to validate the user who sent the authentication request is authorized to access the services of the VC issuer **530**. The identification information **610** may also include additional ID information **623** as needed to further verify the identity of the requesting user. In the current embodiment, it is desirable that the identification information **610** only require the smallest amount of information possible to authenticate that the authentication request **600** is from a known or trusted source so that the control of personal information that is core to the use of a DID is maintained as much as possible.

The authentication request **514** also includes the ID value **512**. As discussed previously, the ID value **512** is generated using the private key **506**. The authentication request also includes request information **620**. The request information **620** includes information that indicates that the digital wallet **510** wants to receive an authentication token from the ID provider **520**.

As shown at **501** in FIG. 5, the digital wallet **510** provides the authentication request **514** to the ID provider **520**. The ID provider **520** includes an authenticator **521**. In operation, the authenticator **521** uses the identification information (i.e., identification information **610**) included in the authentication request to authenticate or validate the user of the digital wallet **510** who provided the authentication request.

Upon a successful authentication or validation, a token generator **522** that is included in the ID provider **520** generates an authentication token **525** that can be presented to the VC issuer **530** for access to one or more verifiable claims **535** as will be explained. In the embodiment, the authentication token **525** includes the ID value **512** that was received as part of the authentication request **514**. In one embodiment, the authentication token **525** may be a JSON document. It will be appreciated that the ID provider **520** will typically have no understanding of the ID value **512** and will not have the ability to interpret or otherwise validate the ID value **512**. In other words, the ID provider **520** will not decrypt the ID value **512**. Rather, the token generator **522** is configured to simply include the ID value **512** in the authentication token **525** for later use by the VC issuer **530**.

FIG. 7 illustrates a specific embodiment of the authentication token **525**. As illustrated, the authentication token **525**

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includes authentication proofs **710**. The authentication proofs **710** include one or more claims issued by the ID provider **520** that verify that the DID owner **201** of the digital wallet **510** is a trusted party that should be given access to the services of the VC issuer **530**. The type and format of the authentication proofs **710** may be dependent on the type of protocol of the ID provider **520**.

The authentication token **525** also includes the ID value **512**. As will be explained, the ID value **512** is used to ensure that a request for the one or more verifiable claims **535** is made by the same user as the user authenticated or validated by the ID provider **520**, which in the embodiment is the DID owner **201**. In some embodiments, the authentication token **525** also includes a signature **720**. The signature **720** is typically a cryptographic mechanism (such as a digital signature) that is used to detect whether the authentication token has been tampered with since the time it was issued and can be used to validate that authentication token **525** was generated by the ID provider **520**. Thus, in embodiments including the signature **720**, the ID provider **520** would use a private key to generate the signature **720** and then the VC issuer **530** would use a corresponding public key to validate the signature. In other embodiments, the signature **720** may be omitted.

As shown at **502** in FIG. 5, the ID provider **520** provides the authentication token **525** to the digital wallet **510**. The digital wallet **510** includes a VC requestor **515**. In operation, the VC requestor **515** is configured to generate a VC request **516** that is a request to the VC issuer **530** for one or more verifiable claims **535** that are to be associated with the DID **205**. The VC request may also include the DID **205** so that the VC issuer **530** knows which DID to associate the credentials with.

FIG. 8 illustrates a specific embodiment of the VC request **516**. As illustrated, the VC request includes the authentication token **525** which has been generated to include the ID value **512** as previously described. The VC request may also include the DID **205** as also previously described.

The VC request **516** includes VC request information **810**. The VC request information **810** is configured to specify the one or more verifiable claims **535** that are desired by the DID owner **201** of the digital wallet **510**. For example, if the VC issuer **530** is a government entity such as the DMV, then the VC request information **810** may specify that verifiable claims related to a driver's license or driving history are desired. Alternatively, if the VC issuer **530** is an employer, then the VC request information **810** may specify that verifiable claims related to employment history are desired.

In some embodiments, the VC request **516** also includes a signature **820**. The signature **820** is typically a cryptographic mechanism (such as a digital signature) that is generated using the private key **206** associated with the DID **205**. As with the signature **720** discussed previously, the signature **820** can be used to detect if the VC request **516** has been tampered with and can also be used to verify that the VC request **516** has been generated by the digital wallet **510**.

As shown at **503** in FIG. 5, the digital wallet **510** provides the VC request **516** to the VC issuer **530**. The VC issuer **530** includes an authenticator **531**. In operation, the authenticator **531** is configured to validate the signature **820**. In some embodiments, the validation includes validating the signature **820** by use of the public key **207** associated with the DID **205**. In addition, the authenticator **531** is configured to validate the authentication token **525** to ensure that the ID provider properly generated the token **525**. In some embodiments, the validation includes validating the signature **720** by use of a public key associated with the ID provider **520**.

In addition, the authenticator **531** is configured to validate the ID value **512** to ensure that the authentication token **525** is being provided by the DID owner **201** of the DID **205** using the digital wallet **510**. As mentioned above, in one embodiment the ID value **512** is a randomized value or hash generated using the private key **206**. In such embodiment, the authenticator **531** accesses the DID **205** from the VC request **516** and uses the DID to access the distributed ledger (i.e., distributed ledger **220**) to obtain the public key **207** associated with DID **205** from the DID document **210** of the DID **205** in the manner previously described. In other embodiments, the VC issuer **530** may already have access to the DID **205** and thus does not need to access it from the VC request **516** when using the DID **205** to access the public key **207** from the distributed ledger. Alternatively, the VC issuer **530** may already have access to the public key **207** associated with the DID **205** from a prior interaction with the DID owner **201**.

Once the public key **207** associated with the DID **205** has been accessed, the authenticator **531** uses the public key **207** to validate the ID value **512**. For example, since the ID value **512** was generated using the private key **206** as previously discussed, the authenticator **531** is able to use the public key **207** to decrypt the ID value **512** to ascertain that the ID value **512** was signed by the same entity that signed the DID **205**. In other words, the authenticator **531** determines if the DID **205** and the ID value **512** were both signed using the private key **206**. If the authenticator **531** determines that the DID **205** and the ID value **512** were both signed using the private key **206**, the authenticator **531** will validate the ID value **512** and allow the VC issuer **530** to issue the requested verifiable claims.

As also mentioned above, in one embodiment, the ID value **512** is a hash of the DID. In such embodiment, the authenticator **531** makes a comparison of a hash of the DID **205** with the hash of the DID that comprises ID **512**. If the comparison shows that the hash of the DID that comprises the ID value **512** matches the hash of the DID **205** possessed or generated by the authenticator **531**, the authenticator **531** will validate the ID value **512** and allow the VC issuer **530** to issue the requested verifiable claims.

Likewise, in the embodiment where the ID value is the DID **205**, the authenticator **531** compares the DID **205** included in the VC request **516** with the DID that comprises the ID value **205**. If the comparison shows that the DID **205** that comprises the ID value **512** matches the DID **205**, the authenticator **531** will validate the ID value **512** and allow the VC issuer **530** to issue the requested verifiable claims. In both the embodiment where the hash of the DID is included in the VC request and in the embodiment where the DID itself is included in the VC request, since the authenticator **531** validates the signature **820** in the manner previously described, the VC issuer **530** has assurance that the DID **205** used to sign the signature **820** is the same DID that is hashed or that is included in the VC request and that the DID **205** is owned by the DID owner **201**.

Of course, if the ID value **512** is not included in the VC request **516** or if the authenticator **531** fails to validate the ID value **512**, then the VC issuer **530** will not issue the requested verifiable claims. Thus, the use of the ID value **512** ensures that the authentication proofs **710** included in the authentication token **525** are bound to the DID **205** so that no other DID can use the authentication proofs **710** to gain authorization to receive the requested verifiable claims.

Accordingly, if an entity other than the DID owner **201** has in some malicious manner hijacked the authentication token **525** and tries to associate this token with his or her

own DID to thus receive the verifiable credentials from the VC issuer **530**, the process will fail. That is, the DID of the malicious party and the ID value **512** will not have been signed by the same private key and thus the authenticator **531** will not authenticate the ID value **512**. In such cases, the VC issuer **530** will not provide the requested verifiable claims.

In some embodiments, the DID owner **201** may try to use the authentication token **525** to receive verifiable claims for a second DID he or she owns such as DID **331**. However, since the ID value **512** will have been used in relation to the DID **205** as explained, the authenticator **531** will reject the request since the ID value **512** can only be used in relation to the DID **205**. Thus, the DID owner **201** is required to generate a new ID value for the DID **331** and then request a new authentication token that includes the new ID value to request the verifiable claims for the DID **331**. This provides a further level of security as it prevents a malicious party from spoofing a DID owned by the DID owner **201** and using an existing authentication value on behalf of the spoofed DID.

As previously described, in some embodiments the VC request **516** may include the signature **820**. In such embodiments, the authenticator **531** also uses the public key **207** to validate the signature **820**. This provides a further level of security as if the signature **820** is not validated due to tampering with the VC request, the authenticator **531** can prevent the verifiable claims from being provided to the digital wallet **510**.

The VC issuer **530** also includes a VC generator **532** that is configured to generate one or more verifiable claims **535** that are associated with the DID **205**. The verifiable claims will be described in more detail in relation to FIGS. **9A** and **9B**.

FIG. **9A** illustrates an example data structure that represents a claim **910**. The claim **910** includes a subject **911**, a property **912** and a value **913**. For example, the subject **911** corresponds to the DID owner **201**. The DID **205** is recorded as part of the subject **911**. The property **912** may be any property of the DID owner **201**, such as a name, a phone number, an email address, etc. The value **913** is the value of the corresponding property **912**. For example, when the property is "name", the value would be the name of the owner of the DID, e.g., John Doe; when the property is "phone number", the value would be the phone number of the owner of the DID, e.g., 1-800-123-4567.

FIG. **9B** illustrates an example data structure of a verifiable claim or credential **535**. The verifiable claim includes the claim **910** of FIG. **9A** and the DID **205**. The verifiable claim **535** also includes a signature **930**, which is generated by signing the verifiable claim **535** by a private key of the issuer, in this case the private key of the VC issuer **530**. The signature **930** is typically a cryptographic mechanism (such as a digital signature) that is used to detect whether the verifiable claim **535** has been tampered with since the time that the verifiable claim was issued and can be used to verify the identity of the VC issuer **530**.

Once the verifiable claim **535** is generated, at least a portion of data related to the verifiable claim **535** is propagated onto a distributed ledger (e.g., the distributed ledger **220**), such that a relying entity can use the portion of data propagated onto the distributed ledger to verify the verifiable claim. In some embodiments, the public key corresponding to the private key of VC issuer **530** is also propagated onto the distributed ledger. In some embodiments, a hash of the public key or a hash of the verifiable claim or claim **535** is propagated onto the distributed ledger.

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In some embodiments, the verifiable claim **535** also includes various metadata **920** related to the verifiable claim **535**. For example, the metadata includes, but is not limited to, (1) a unique identifier **921** identifying the corresponding verified claim or credential, (2) one or more conditions **922** for accessing the verifiable claim or credential, or (3) duration information metadata **923** related to a duration of time that the VC issuer **530** wants the verifiable claim **535** to be valid for or useable for.

The one or more conditions metadata **922** for accessing the verifiable claim **535**, include but are not limited to, (1) requiring the relying entity to pay a predetermined amount of cryptocurrency or type of currency, (2) requiring the relying entity to provide identification information, (3) requiring the relying entity to provide one or more verifiable claims, (4) requiring the relying entity to grant permission for accessing a portion of data, and/or (5) requiring the relying entity to provide a particular service.

The duration information metadata **923** includes, but is not limited to, (1) an expiration time of the corresponding verifiable claim **535**, (2) a predetermined number of times that the corresponding verifiable claim **535** can be accessed or used, (3) a mechanism that automatically causes the verifiable claim **535** to expire in response to a directive from the VC issuer **530**, or (4) a mechanism that allows the DID owner **201** to manually cause the verifiable claim **535** to expire.

As shown at **504**, the VC issuer **504** provides the one or more verifiable claims **535** to the digital wallet **510** for the use of the DID owner **201**. The one or more verifiable claims **535** may be stored in a VC store **517** that is part of the digital wallet **510**. In addition to or alternatively, the one or more verifiable claims **535** may be stored in one or more of the ID hubs **411-414** or may be stored in both the VC store **517** and the one or more ID hubs **411-414**.

The one or more verifiable claims **535** can subsequently be used by the DID owner **201** in interacting with third party entities that require the verifiable claims before providing a service to the DID owner. For example, if the VC issuer **530** is a government entity such as the DMV, then the one or more verifiable claims **535** may be related to a driver's license or driving history. At a subsequent time to receiving the one or more verifiable claims related to a driver's license or driving history, the DID owner **201** can present these verifiable claims to a car rental agency when renting a car. The car rental agency can use the verifiable claims to determine if the DID owner **201** has a valid driver's license and/or if he or she is a safe driver when determining whether to rent the car or not to the DID owner **201**.

FIG. **10** illustrates a more detailed process flow **1000** for the interactions between the digital wallet **510**, the ID provider **520** and the VC issuer **530** and their interactions with the distributed ledger **220**. It will be appreciated that there are no temporal limitations to the process flow **1000** as some of the interactions may occur at a time earlier than a subsequent interaction and some of the interactions may occur one after the other in a short time frame.

As shown at interaction **1001**, the digital wallet **510** accesses the distributed ledger **220** when generating the DID **205** using the private key **206** and records the DID document **210** or a pointer thereto onto the distributed ledger. Interaction **1002** represents that the DID **205** has been tied to the block chain and that the public key **207** is available for use by entities that are given access to the public key. Interactions **1001** and **1002** also represents that the digital wallet **510** has generated the ID value **512**. In one embodiment, generating the ID value includes using the private key **206**

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so that the ID value **512** can be decrypted using the public key **207** that is accessible via the distributed ledger **220**. In other embodiments, generating the ID value includes using a hash of the DID or the DID itself as the ID value.

During interaction **1003**, the digital wallet **510** provides the authentication request **514** including the ID value **512** to the ID provider **520**. During interaction **1004**, the ID provider **520** validates the authentication request **514** and generates the authentication token **525** upon a successful validation. The ID provider returns the authentication token **525** including the ID value **512** to the digital wallet **510** during interaction **1005**.

During interaction **1006**, the digital wallet **510** generates the VC request **516** that includes the authentication token **525** and the ID value **512**. The VC request **516** is sent to the VC issuer **530** during interaction **1007**.

During interaction **1008**, the VC issuer **530** validates the authentication token **525**. In one embodiment, during interaction **1009**, the VC issuer **530** accesses the distributed ledger **220** to thereby access the public key **207**. Interaction **1010** represents that the public key **207** has been accessed by the VC issuer **530**. In other embodiments, the interactions **1009** and **1010** may not be required.

During interaction **1011**, the VC issuer **530** validates the ID value **512**, which in one embodiment includes using the public key **207**. Upon successful validation, the verifiable claims **535** are returned to the digital wallet **510** during interaction **1012**.

The following discussion now refers to a number of methods and method acts that may be performed. Although the method acts may be discussed in a certain order or illustrated in a flow chart as occurring in a particular order, no particular ordering is required unless specifically stated, or required because an act is dependent on another act being completed prior to the act being performed.

FIG. **11** illustrates a flow chart of an example method **1100** for a computing system of a digital wallet to receive an authentication token from an identity provider, the authentication token being used by the computing system in a request for one or more verifiable claims from a verifiable claim issuer so that the computing system can be validated by the verifiable claim issuer, the computing system being associated with a Decentralized Identifier (DID) that is backed by a distributed ledger. The method **1100** will be described with respect to one or more of FIGS. **2-10** discussed previously.

The method **1100** includes generating an identification value associated with a DID of a DID owner (**1110**). For example, as previously described the ID value generator **511** of the digital wallet **510** generates the ID value **512**. In one embodiment, the ID value **512** is generated using the private key **206** that is associated with the DID **205** of the DID owner **201**. The private key **206** has a corresponding public key **207** that is accessible via the distributed ledger **220**. In such embodiment, the ID value **512** may be a cryptographic random number or hash that is signed by the private key **206**. In another embodiment, the ID generator **511** is configured to generate a hash of the DID **205** as the ID value **512**. In a further embodiment, the ID generator is configured to use the DID **205** as the generated ID value **512**.

The method **1100** includes generating a first request for an authentication token from an identification provider, the authentication token being configured to authenticate the computing system with a verifiable claim issuer, the first request including the identification value (**1120**). For example, as previously described the authentication requestor **513** of the digital wallet generates the authentication

tion request **514** to request an authentication token **525** from the ID provider **520**. The authentication request **514** includes the ID value **512**. In some embodiments, the authentication request also includes identification information such as a username and password and request information that requests the authentication token.

The method **1100** includes providing the first request to the identification provider (**1130**). For example, as previously described the digital wallet **510** provides the authentication request **514** to the ID provider **520**.

The method **1100** includes receiving, in response to the identification provider validating the first request, an authentication token including the identification value from the identification provider (**1140**). For example, as previously described the ID provider **520** may validate the authentication request **514**. Upon a successful validation, the ID provider **520** generates the authentication token **525** and includes the ID value **512** in the token. The authentication token is then received by the digital wallet **510**.

The method **1100** includes generating a second request for one or more verifiable claims from the verifiable claim issuer, the second request including the DID and authentication token including the identification value (**1150**). For example, as previously described the VC requestor **515** of the digital wallet **510** generates the VC request **516**. The VC request **516** includes the authentication token **525** and the ID value **512**. In some embodiments, the VC request **516** may also include request information that specifies the desired verifiable claims and a cryptographic signature that is used for validating the VC request.

The method **1100** includes receiving, in response to the verifiable claim issuer validating the authentication token and the identification value, one or more verifiable claims from the verifiable claim issuer (**1160**). For example, as previously described the VC issuer **530** validates the authentication token **525** and then validates the ID value **512** using the public key **207** that is accessed via the distributed ledger **220**. Upon a successful validation, the VC issuer **530** generates the one or more verifiable claims **535**, which are then received by the digital wallet **510** and stored in one or both of the VC store **517** and the identity hubs **411-414** for future use.

FIG. **12** illustrates a flow chart of an example method **1200** for a computing system of an identity provider to generate an authentication token for use by a digital wallet in a request for one or more verifiable claims from a verifiable claim issuer so that the digital wallet can be validated by the verifiable claim issuer, the digital wallet being associated with a Decentralized Identifier (DID) that is backed by a distributed ledger. The method **1200** will be described with respect to one or more of FIGS. **2-10** discussed previously.

The method **1200** includes receiving an authentication request for an authentication token from a digital wallet that is associated with a DID of a DID owner, the authentication token being configured to authenticate the digital wallet with a verifiable claim issuer, the authentication request including an identification value that is associated with the DID of a DID owner that was generated by the digital wallet (**1210**). For example, as previously described the ID provider **520** receives the authentication request **514** from the digital wallet **510**. The authentication request includes the ID value **512** and requests that the authentication token **525** be generated so that the digital wallet can be authenticated by the VC issuer **530**.

The method **1200** includes validating the authentication request (**1220**). For example, as previously described the authenticator **521** of the ID provider **520** validates the authentication request **514**.

The method **1200** includes, in response to validating the authentication request, generating the authentication token including the identification value (**1230**). For example, as previously described upon a successful validation of the authentication request, the token generator **522** of the ID provider **520** generates the authentication token **525** that includes the ID value **512**.

The method **1200** includes providing the authentication token to the digital wallet (**1240**). For example, as previously described the ID provider **520** provides the authentication token **525** including the ID value **512** to the digital wallet **510**.

FIG. **13** illustrates a flow chart of an example method **1300** for a computing system of a verifiable claim issuer to provide one or more verifiable claims to a digital wallet, the computing system receiving an authentication token issued by an identity provider that validates that the digital wallet is authorized to receive the one or more verifiable claims, the digital wallet being associated with a Decentralized Identifier (DID) that is backed by a distributed ledger. The method **1300** will be described with respect to one or more of FIGS. **2-10** discussed previously.

Method **1300** includes receiving a request for one or more verifiable claims from a digital wallet that is associated with a DID of a DID owner, the request including the DID and an authentication token generated by an identity provider, the authentication token including an identification value that was generated by the digital wallet and that is bound to the DID (**1310**). For example, as previously described the VC issuer **530** receives the VC request **516** from the digital wallet **510** that is associated with the DID **205** of the DID owner **201**. The VC request **516** includes the authentication token **525** that includes the DID **205** and the ID value **512** generated by the digital wallet **510**. The VC request **516** also specifies which verifiable claims the DID owner **201** desires to receive from the VC issuer.

The method **1300** includes validating the authentication token to ensure that the authentication token was issued by the identity provider (**1320**). For example, as previously described, the authenticator **531** of the VC issuer **530** validates the authentication token **525**.

The method **1300** includes validating the identification value (**1330**). For example, as previously described the authenticator **531** of the VC issuer **530** validates the ID value **512**. In one embodiment, the authenticator **531** uses the public key **207** to validate the ID value **512**. In such embodiment, since the ID value **512** was generated and signed using the private key **206**, the public key **207** is able to decrypt the ID value. In another embodiment, the authenticator **531** makes a comparison of a hash of the DID **205** with the hash of the DID that comprises ID **512**. If the comparison shows that the hash of the DID that comprises the ID value **512** matches the hash of the DID **205** possessed or generated by the authenticator **531**. In a further embodiment, the authenticator **531** compares the DID **205** included in the VC request **516** with the DID that comprises the ID value **205**. If the comparison shows that the DID **205** that comprises the ID value **512** matches the DID **205**, the authenticator **531** will validate the ID value **512** and allow the VC issuer **530** to issue the requested verifiable claims. If the ID value **512** is not present or if the ID value is an ID value that is different from the ID value **512**, then the validation will fail.

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The method **1300** includes, in response to validating the authentication token and the identification value, providing one or more verifiable claims to the digital wallet (**1340**). For example, as previously described upon a successful validation of the authentication token **525** and the ID value **512**, the VC generator **532** of the VC issuer **530** generates the one or more verifiable claims **535** according to verifiable claims requested in the VC request **516**. The VC issuer **530** then provides the one or more verifiable claims **535** to digital wallet **510**.

For the processes and methods disclosed herein, the operations performed in the processes and methods may be implemented in differing order. Furthermore, the outlined operations are only provided as examples, and some of the operations may be optional, combined into fewer steps and operations, supplemented with further operations, or expanded into additional operations without detracting from the essence of the disclosed embodiments.

The present invention may be embodied in other specific forms without departing from its spirit or characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

What is claimed is:

1. A method for a computing system of a digital wallet to receive an authentication token from an identity provider, the authentication token being used by the computing system in a request for one or more verifiable claims from a verifiable claim issuer so that the computing system can be validated by the verifiable claim issuer, the computing system being associated with a Decentralized Identifier (DID) that is backed by a distributed ledger, the method comprising:

generating an identification value associated with a DID of a DID owner;

generating a first request for an authentication token from an identity provider, the authentication token being configured to authenticate the digital wallet with the verifiable claim issuer, the first request including the identification value;

provide the first request to the identification provider;

receive, in response to the identification provider validating the first request, the authentication token including the identification value from the identification provider;

generate a second request for one or more verifiable claims from the verifiable claim issuer, the second request including the DID and authentication token including the identification value; and

receive, in response to the verifiable claim issuer validating the authentication token and the identification value, the one or more verifiable claims from the verifiable claim issuer.

2. The method of claim 1, wherein the identification value is a random number or hash value that is signed using a private key associated with the DID, the private key having an associated public key that is accessible via a distributed ledger.

3. The method of claim 2, wherein the associated public key is accessible by accessing a DID document associated with the DID that is tied to the distributed ledger.

4. The method of claim 1, wherein the identification value is a hash of the DID.

5. The method of claim 1, wherein the identification value comprises the DID.

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6. The method of claim 1, wherein the first request includes identification information related to the DID owner and information requesting the authentication token.

7. The method of claim 1, wherein the authentication token includes one or more authentication proofs that are generated by the identity provider.

8. The method of claim 1, wherein the authentication token is cryptographically signed by the identity provider.

9. The method of claim 1, wherein the second request includes information specifying one or more verifiable claims that the DID owner desires to receive from the verifiable claim issuer.

10. The method of claim 1, wherein the second request is cryptographically signed using the private key that was used to generate the identification value.

11. The method of claim 1, wherein the one or more verifiable claims received from the verifiable claim issuer are stored at the computing system and/or stored at an identity hub owned by the DID owner.

12. A method for a computing system of an identity provider to generate an authentication token for use by a digital wallet in a request for one or more verifiable claims from a verifiable claim issuer so that the digital wallet can be validated by the verifiable claim issuer, the digital wallet being associated with a Decentralized Identifier (DID) that is backed by a distributed ledger, the method comprising:

receiving an authentication request for an authentication token from a digital wallet that is associated with a DID of a DID owner, the authentication token being configured to authenticate the digital wallet with the verifiable claim issuer, the authentication request including an identification value associated with the DID that was generated by the digital wallet;

validating the authentication request;

in response to validating the authentication request, generating the authentication token including the identification value; and

providing the authentication token to the digital wallet.

13. The method of claim 12, wherein the authentication request includes identification information related to the DID owner and information requesting the authentication token.

14. The method of claim 12, wherein the authentication token includes one or more authentication proofs that are generated by the computing system.

15. The method of claim 12, wherein the authentication token is cryptographically signed by the computing system.

16. The method of claim 12, wherein the computing system includes the identification value with the authentication token without decrypting the identification value.

17. A method for a computing system of a verifiable claim issuer to provide one or more verifiable claims to a digital wallet, the computing system receiving an authentication token issued by an identity provider that validates that the digital wallet is authorized to receive the one or more verifiable claims, the digital wallet being associated with a Decentralized Identifier (DID) that is backed by a distributed ledger, the method comprising:

receiving a request for one or more verifiable claims from a digital wallet that is associated with a DID of a DID owner, the request including the DID and an authentication token generated by an identity provider, the authentication token including an identification value that was generated by the digital wallet and that is bound to the DID;

validating the authentication token to ensure that the authentication token was issued by the identity provider;

validating the identification value; and

in response to validating the authentication token and the identification value, providing the one or more verifiable claims to the digital wallet. 5

18. The method of claim **17**, wherein the identification value is a random number or hash value that is signed using a private key associated with the DID, wherein validating the identification token comprises using a public key that is associated with the private key and this is accessible via a distributed ledger. 10

19. The method of claim **17**, wherein the identification value is a hash of the DID, wherein validating the identification value comprises determining that the hash of the DID comprising the ID value matches a hash of the DID. 15

20. The method of claim **17**, wherein the identification value is a the DID, wherein validating the identification value comprises determining that the DID comprising the ID value matches the DID. 20

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